

By e-mail

**GOVERNMENT OF INDIA  
MINISTRY OF MINES  
INDIAN BUREAU OF MINES  
OFFICE OF THE REGIONAL CONTROLLER OF MINES, AJMER**

**No. File No. E13843-MCDR- MPCOLST/14/2025-AJM-  
IBM\_RO\_AJM**

Dt : 16/06/2025

Shri/M/s. Suncity Chemical and Minerals Private Limited ,  
27, A-68 Adarsh Society, ITI Rd, Light Industrial Area, Sector G, Shastri Nagar, Jodhpur Jodhpur

**Sub** Approval of Mining Plan along-with Progressive Mine Closure Plan in respect of TKSB18 Limestone Block over an area of 4.80 Ha.  
: situated near village Haripura of Tehsil Kheenvsar in District Nagaur of Rajasthan State, submitted under Rule 16(1) of MCR, 2016.

Ref: 1) Your reference No. online submission of Mining Plan in respect of aforesaid block dated 01/05/2025.

2) Online refer back of document through MPAS portal dated 20/05/2025.

3) Online submission of final copy of mining plan in respect of aforesaid block dated 12/06/2025.

Sir,

In exercise of the powers conferred by clause (b) of sub-section (2) of section 5 of the Mines & Minerals (Development & Regulation) Act, 1957 and in accordance with clause (3) of Rule 16 & clause (1) of Rule 16 of the Minerals (Other than Atomic and Hydro Carbons Energy Minerals) Concession Rules, 2016 read with Government of India Order No.S.O.445(E) dated 28.04.1987 and S.O. 1857(E)/1872(E)dated 18th May, 2016; I hereby **APPROVE** the above said Mining Plan along with Progressive Mine Closure Plan (PMCP) **for 05 years period from the date of execution of the Mining Lease**. This approval is subject to the following conditions:

1. This Mining Plan (including PMCP) is approved without prejudice to any other laws applicable to the mine / area from time to time whether made by the Central Government, State Government or any other Authority and any order or direction from any court of competent jurisdiction.
2. It is also clarified that the approval of this Mining Plan (including PMCP) does not in any way imply the approval of the Government in terms of any other provisions of the Mines and Minerals (Development & Regulation) Act, 1957 as amended or the rules framed there under and any other laws including Environment (Protection) Act, 1986, and the rules framed there under.
3. It is further clarified that the approval of this Mining Plan (including PMCP) is subject to the provision of Forest (Conservation) Act, 1980, Forest Conservation Rules, 2003 and other relevant statutes, order and guidelines as may be applicable to the lease area from time to time.
4. The provisions of the Mines Act, 1952 and Rules & Regulations made there under shall be complied with.
5. The execution of Mining Plan shall be subjected to vacations of prohibitory orders /notices, if any.
6. If anything is found to be concealed as required under the Mines Act in the contents of the Mining Plan and the proposal for rectification has not been made, the approval shall be deemed to have been withdrawn with immediate effect.
7. This department does not undertake any responsibility regarding correctness of the boundaries of the lease area shown on the ground.
8. This approval is restricted for mining operation and mining related activities within the mining lease area only. The mining lease area is as shown on the statutory plans by the Lessee/QP/Applicant and Indian Bureau of Mines has not undertaken verification of mining lease boundary on the ground.
9. Your attention is invited to the Supreme Court interim order in W.P.(C) No.202 dated 12.12.1996 for compliance. The Approval of this Mining Plan (including PMCP) is therefore, issued without prejudice to and is subject to the said directions of the Supreme Court as applicable.
10. The approval of this Mining Plan with PMCP is subject to condition that you shall maintain boundary pillars as indicated in Rule 12 (v) of Minerals (Other than Atomic and Hydro Carbons Energy Minerals) Concession Rules, 2016.
11. At any stage, if it is observed that the information furnished in the document are incorrect or misrepresent facts, the approval of the document shall be revoked with immediate effect.
12. The inventory of mineral/ore stock above the limit prescribed in the threshold values of minerals and below the cutoff grade shall be maintained in a bound register indicating the quantity and quality of material stacked. The month wise inventory of such materials shall be updated, and the annual summary shall be submitted to Regional Controller of Mines, IBM, Ajmer before 30<sup>th</sup> June of every year.
13. The lessee shall ensure grassing/re-grassing in the reclaimed land in the lease area in accordance with the Hon'ble Supreme Court

judgement dated 08.01.2020.

14. It shall be mandatory for the project proponent, abstracting ground water, to obtain No Objection Certificate from Central Water Authority or, the concerned State/Union Territory Ground Water Authority, as the case may be.
15. The applicant shall complete detailed exploration (G1 level) and prepare a detailed feasibility study report conforming to Part IV and V of the Mineral (Evidence of Mineral Contents) Rule 2015 over the entire area under the mining lease, within a period of five years from the date of commencement of such mining lease.
16. Latest Land ownership details and khasra plan duly verified/ authenticated from State DMG shall be submitted to this office within six months from date of approval of this document.
17. All the technical studies like ground vibration, bulk density etc. shall be conducted within six months from the date of commencement of mining operation and report of the same to be submitted to this office.
18. This approval is given **for 05 years period from the date of execution of the Mining Lease** for the mining activities to be carried out within the mining lease area.

Yours faithfully,

**(Chandresh Bohra)**

**Regional Controller of Mines**

**Regional Office, Ajmer**

**Indian Bureau of Mines**

Copy without enclosure forwarded for information to

Shri NimishSinghvi, Qualified Person, 15, New Glass Factory Colony, Sunderwas, Udaipur,(Rajasthan)

**Regional Controller of Mines**

**Regional Office, Ajmer**

**Indian Bureau of Mines**

N.O.O.: Copy forwarded for kind information to:

- (i) The Controller of Mines (NZ), Indian Bureau of Mines, Udaipur.
- (ii) The Director, Department of Mines & Geology, Government of Rajasthan, Shastri Circle, Udaipur

**Regional Controller of Mines**

**Regional Office, Ajmer**

**Indian Bureau of Mines**

Chapter 1 : General Information

1.1 : Lease Details

|                           |                                                                                                     |
|---------------------------|-----------------------------------------------------------------------------------------------------|
| IBM Registration Number : | IBM/47130/2025                                                                                      |
| Lease Code :              | PB216                                                                                               |
| Mine Code :               | Nil                                                                                                 |
| Name of Lessee :          | Suncity Chemical And Minerals Private Limited                                                       |
| Address of Lessee :       | 27, A-68 Adarsh Society, ITI Rd, Light Industrial Area, Sector G, Shastri Nagar, Jodhpur<br>Jodhpur |
| Type of Lessee :          | Private                                                                                             |
| Name of Mining Lease :    | Haripura                                                                                            |
| State :                   | RAJASTHAN                                                                                           |
| District :                | NAGAU                                                                                               |
| Tehsil/ Taluk/ Mandal :   | Kheenvsar                                                                                           |
| Village :                 | Haripura                                                                                            |
| Lease Area (Ha) :         | 4.80                                                                                                |
| Forest Area (Ha) :        | 0.0000                                                                                              |
| Name of Minerals :        | LIMESTONE                                                                                           |
|                           |                                                                                                     |

|                                    |             |
|------------------------------------|-------------|
| Name of associated minerals :      |             |
| Type :                             | Fresh Grant |
| Period of the proposal (FY) from : | Nil         |
| Period of the proposal (FY) to :   | Nil         |
| Type of working :                  | Opencast    |
| Nature of Use :                    | Non Captive |
| Category of Mine :                 | Category A  |

1.1.1 : Initial/subsequent Lease grant details

|       |      |     |                           |                         |
|-------|------|-----|---------------------------|-------------------------|
| Grant | From | To  | Lease deed execution date | Lease registration date |
| Nil   | Nil  | Nil | Nil                       | Nil                     |

1.1.2 : Mining Plan Submission Criteria Details

|                    |                                                                                                                                                                                                               |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Type of Document : | Mining Plan Under Rule 16(1) Of MCR 2016                                                                                                                                                                      |
| LOI Number :       | P.2(9)MINE/GROUP-2/2024 , With this LOI we are submitting the mining plan and after getting all the statutory clearance and allocation of mines lease we will start the mining operation as per approved plan |
| Date :             | 22/11/2024                                                                                                                                                                                                    |

1.2 : Land Ownership Details

|                                   |                                                 |
|-----------------------------------|-------------------------------------------------|
| View Land Ownership Details Excel | <a href="#">Land_Ownership_Details (5).xlsx</a> |
|-----------------------------------|-------------------------------------------------|

1.3 : Existing Lease

|                     |     |
|---------------------|-----|
| Date of Execution : | Nil |
|---------------------|-----|

1.3.1 : Approval of earlier Mining Plan & Its Subsequent Review in Chronological Order

| S.N. | Letter Number | Date | Period |     | Type Of Approved Document | Remark |
|------|---------------|------|--------|-----|---------------------------|--------|
|      |               |      | From   | To  |                           |        |
| 1    | Nil           | Nil  | Nil    | Nil | Nil                       | Nil    |

1.3.2 : Partial Surrenderd Area During Stages of Operations in Chronological Order

Not Applicable

1.3.3 : Transfer of Lease Area Subsequent to Grant

Not Applicable

1.3.4 : Statutory Compliances

1.3.4.1 : Environment Clearance

|              |     |
|--------------|-----|
| Applicable : | No  |
| Letter No :  | Nil |
| Date :       | Nil |

|               |     |
|---------------|-----|
| Validity :    | Nil |
| ROM Mineral : | Nil |

1.3.4.2 : SPCB Approvals

|               |     |
|---------------|-----|
| Letter No :   | Nil |
| Approval of : | Nil |
| Date :        | Nil |
| Validity :    | Nil |
| ROM Mineral : | Nil |

1.3.4.3 : Forest Clearance

|              |     |
|--------------|-----|
| Applicable : | No  |
| Letter No :  | Nil |
| Date :       | Nil |
| Validity :   | Nil |
| Area (Ha) :  | Nil |

1.3.4.4 : Land Acquisition Details

|                                 |        |
|---------------------------------|--------|
| Total Area Acquired in hectare: | 0.0000 |
| Total Amount Paid (INR) :       | 0.0000 |



1.3.5 : Mine Location Details

|                    |        |
|--------------------|--------|
| Toposheet Number : | 45 F/5 |
|--------------------|--------|

1.3.5.1 : Location of Boundary Pillars

|                                         |     |
|-----------------------------------------|-----|
| View Location of Boundary Pillars Excel | Nil |
|-----------------------------------------|-----|

1.3.6 : Owner/Nominated Owner Details

| Name          | PAN of owner / Nominated Owner | Address of owner/ Nominated Owner        | Mobile / Contact Number | Email                   | Please attach Minutes of Board Resolution in case of Nominated Owner |
|---------------|--------------------------------|------------------------------------------|-------------------------|-------------------------|----------------------------------------------------------------------|
| Mehul Agarwal | CXSPA7422D                     | D17 Shastri Nagar Jhodhpur<br>Sadiya Raj | 9414023531              | mehul@suncitysheets.com | <a href="#">POA with BR and list of directors.pdf</a>                |

1.3.7 : Qualified Person Details as per M(OAHCEM)CR, 2016

| S.N. | Prefix | Name          | PAN of QP  | Address                                                       | Mobile no. | Qualification | Exp in years as prescribed under the rule | Email                    |
|------|--------|---------------|------------|---------------------------------------------------------------|------------|---------------|-------------------------------------------|--------------------------|
| 1    | Mr     | NimishSinghvi | AGNPS9913R | 15, NewGlass<br>FactoryColony,<br>Sunderwas,<br>Udaipur(Raj.) | 8209042477 | BE Mining     | 5                                         | nimesh.singhvi@gmail.com |

Chapter 2A : Geology & Exploration

2A.1 : Geology

2A.1.1 : Topography

|                                         |           |
|-----------------------------------------|-----------|
| Terrain :                               | Plain     |
| Highest Level (m) from MSL :            | Nil       |
| Lowest Level (m) from MSL :             | Nil       |
| Average Level (m) from MSL :            | 312.0000  |
| Drainage Pattern :                      | Dendritic |
| Order of Stream :                       | Order 1   |
| Min Dist of Stream from Lease Area(m) : | 2500.0000 |

2A.1.2 : Details of Physiographic features and Infrastructures available in and around the lease/ block area

| Description                         | Location if existing Within the lease/block area                       | Distance from boundary periphery in kms, if existing outside the lease/block area. (within 5.00Kms) | Remark if any                                                               |
|-------------------------------------|------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| River/Nallah/Reservoir              | no                                                                     | 2.5                                                                                                 | No perennial river originates or flows from this part of investigated area. |
| Public roads (Tar road, cart road)  | no                                                                     | 0.938                                                                                               | The investigated area is well connected by tar road.                        |
| Railway track                       | no                                                                     | 0                                                                                                   | Nil                                                                         |
| Human settlements                   | Yes (Two House near Pillar T2 And 1 house is between Pillar T1 and S1) | 1.2                                                                                                 | Nil                                                                         |
| Archaeological monuments/ places of | no                                                                     | 0                                                                                                   | Nil                                                                         |



|                                                 |     |      |                                                      |
|-------------------------------------------------|-----|------|------------------------------------------------------|
| worships/public utilities etc                   |     |      |                                                      |
| Wild life sanctuaries/ national parks           | No  | 0    | Nil                                                  |
| Coastal Regulation Zone (CRZ)                   | no  | 0    | Nil                                                  |
| Powertransmision lines/telephone lines          | Yes | 0.13 | One power transmission line is Present in lease area |
| Firing range                                    | no  | 0    | Nil                                                  |
| Ordinance factory                               | no  | 0    | Nil                                                  |
| grazing land/ burial ground or cremation ground | no  | 0    | Nil                                                  |
| Any other specify                               | no  | 0    | Nil                                                  |

|                                   |                                     |
|-----------------------------------|-------------------------------------|
| Particulars                       | Distance from lease boundary in kms |
| Near by village                   | Haripura 0.6 km                     |
| Nearest Railway station           | Nagaur 34.26km                      |
| Nearest Port                      | Jodhpur Airport 90KM                |
| Distance of SH/NH from lease area | NH 62- 3.0Km                        |

2A.1.3 : Regional Geology

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>Regional Geology</b></p> <p>Regional geologically the area belongs to Bilara Group of Marwar Super group. The main lithounits are cherty limestone, Dolomitic limestone and greyish black laminated limestone. These formations are at places overlain by lime kankar bed of recent age or sandstone of Nagaur Group. The Bilara limestone constitutes a prominent N-S trending belt in central Rajasthan that extends for about 150 km in length and 2 to 6 km in width. It is being developed and used for making lime since past five decades. The Bilara Group represents a marine environment of deposition. The limestone is mostly light to dark grey, fine to coarse-grained and massive to laminated. The beds show rolling low dips mostly varying from 1 to 5°. General trend of limestone bed is NE-SW and are almost horizontally disposed.</p> |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

2A.1.4 : Local Geology & Structure

**2A.1.4.1 : Local Geological Set-up**

Geologically the area mainly comprises lith of acies of Gotan formation of Bilara group of Marwar Super group which has been assigned Cambrian age. The main rock types of the investigated area are cherty dolomitic limestone, fine to medium grained grey colour limestone. These lito units are overlained by kankar formation of Quaternary age. The cherty dolomitic limestone is light reddish to dirty white in colour having bouldry nature. The fine to medium grained grey colour lime stone is exposed below the cherty dolomitic limestone, which are intersected in drilling as well as in pits. These limestones are bedded in nature and show horizontal deposition. These rocks are overlained by kankar formation of Quaternary age. In subsurface the other rock units like calcareous red shale, shaly limestone and brecciated dolomitic limestone also intersected.

**2A.1.4.2 : Structure**

The limestone is trending almost N-S with rolling dip.

**2A.1.4.3 : Lithology, Petrographic & Mineralogical Description for Major, Associated & Indicator Minerals**

The major rock types encountered in the area are Limestone, dolomite, a limestone, apart from that there is fine aeolian sand and silt. Dolomitic Limestone The dolomitic limestone of area is earthy white to ash in colour, medium to coarse grained crystalline texture and at top surface characteristic elephant skin weathering is present. Thcherty dolomitic limestone is light reddish to dirty white in colour having boundary nature. Limestone The limestone of area appears to be good quality for high grade limestone and having grey colour, laminated cherty bands and fine to medium grain texture. The quality of the limestone rocks may increase with depth. The fine to medium grained grey colour limestone is exposed below the cherty dolomitic limestone, which are intersected in drilling as well as in pits. These limestones are bedded in nature and show horizontal deposition. These rocks are overlayed by kankar Formation of Quaternary age. In subsurface the other rock units like calcareous red shale, shaly limestone and brecciated dolomitic limestone also intersected.

**2A.1.4.4 : Mode of Occurance & Controls of Mineralization**

Geologically, the area belongs to Bilara Group of Marwar Supergroup. The main rock types in the area are cherty dolomitic limestone, fine to medium grained grey colour limestone. These rocks are overlayed by Kankar formation of Quaternary age .boundary nature havingNE-SWstrike;horizontalinnatureandatplacesdipping1 to5 in SW direction.

**2A.1.4.5 : Extent of Weathering/ Alteration**

Evidence of chemical weathering canbe seen in areas where limestone is exposed to the elements. Rain water erodes the verticaljointsandhorizontalbeddingplanesinlimestone.Indoingthiskarst scenery screated. Dolomitization refers to the varied suggested manners in which a predominantly calciterich calcium bearingse dimentary rock suchas limestone.

|                                   |            |
|-----------------------------------|------------|
| 2A.1.4.6 : Nature/Form of Mineral |            |
| Specify If any other              | 40cm x40cm |

|                                                                                                                                                                                                                                          |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2A.1.4.7 : Extent of Mineralization                                                                                                                                                                                                      |
| The number of Mineral Zones is from 1 to2. The limestone belt has a generalstrike direction inthe areaasN- Shaving almosthorizontalbedsandatplacesdipping1 to5 degrees in SWdirection. TheCumulativethicknessofmineralizedzoneis12.50 m. |

|                                                                       |
|-----------------------------------------------------------------------|
| 2A.1.4.8 : Deposit Type (as per MEMC Rule)                            |
| These limestones are bedded in nature and show horizontal deposition. |

|                                |   |    |    |    |   |    |
|--------------------------------|---|----|----|----|---|----|
| Strike / Trend of the Ore Body |   |    |    |    |   |    |
| NE                             | 0 | SW | to | NE | 0 | SW |

|                                        |                                        |
|----------------------------------------|----------------------------------------|
| Amount of Dip of the Ore Body (degree) | Amount of Dip of the Ore Body (degree) |
| 1                                      | 5                                      |
| (from)                                 | (to)                                   |

|                               |   |    |                                             |                     |     |   |
|-------------------------------|---|----|---------------------------------------------|---------------------|-----|---|
| Dip Direction of the Ore Body |   |    | Plunge of Mineral Body<br>(degree) (if any) | Direction of Plunge |     |   |
| NW                            | 2 | SE | Nil                                         | W                   | Nil | W |

2A.2: Exploration

2A.2.1: Summary of The Previous Exploration (for fresh grant) / During Last Plan Period (for existing leases)

|                                           |
|-------------------------------------------|
| Name of The Agency                        |
| Directorate of Minesand Geology,Rajasthan |

2A.2.1.1: Geological Mapping

| SI.No. | Year       |            | Scale   | Area Covered (Ha) |
|--------|------------|------------|---------|-------------------|
|        | From       | To         |         |                   |
| 1      | 24/01/2020 | 12/02/2020 | 1:10000 | 4.8000            |

2A.2.1.2: Airborne Geophysical Survey

| SI.No. | Type of Survey | Spacing (m) | Total line (km) | Area Covered (Ha) | Latitude (dd:mm:ss.ss) |     | Longitude (dd:mm:ss.ss) |     |
|--------|----------------|-------------|-----------------|-------------------|------------------------|-----|-------------------------|-----|
|        |                |             |                 |                   | Form                   | To  | Form                    | To  |
| 1      | Nil            | Nil         | Nil             | Nil               | Nil                    | Nil | Nil                     | Nil |

2A.2.1.3: Ground Geophysical Survey

| SI.No. | Type of Survey | Spacing (m) | Total line (km) | Area Covered (Ha) | Latitude (dd:mm:ss.ss) |     | Longitude (dd:mm:ss.ss) |     |
|--------|----------------|-------------|-----------------|-------------------|------------------------|-----|-------------------------|-----|
|        |                |             |                 |                   | Form                   | To  | Form                    | To  |
| 1      | Nil            | Nil         | Nil             | Nil               | Nil                    | Nil | Nil                     | Nil |

2A.2.1.4: Geochemical Survey

| SI.No. | Type of Sample | No of Samples | Aanlysis report | Area Covered (Ha) |
|--------|----------------|---------------|-----------------|-------------------|
| 1      | Nil            | Nil           | Nil             | Nil               |

2A.2.1.5: Pitting

|                |  |  |  |  |  |  |  |  |  |  |  |  |  |
|----------------|--|--|--|--|--|--|--|--|--|--|--|--|--|
| Number of Pits |  |  |  |  |  |  |  |  |  |  |  |  |  |
| Nil            |  |  |  |  |  |  |  |  |  |  |  |  |  |

| SI.No. | Year | Pit ID | Length of Pit (m) | Width of Pit (m) | Depth of Pit (m) | Depth (from) | Depth(to) | Running mtr | Litho units | Name of the | Av Grade(in | Latitude (dd:mm:ss.ss) | Longitude (dd:mm:ss.ss) |
|--------|------|--------|-------------------|------------------|------------------|--------------|-----------|-------------|-------------|-------------|-------------|------------------------|-------------------------|
|--------|------|--------|-------------------|------------------|------------------|--------------|-----------|-------------|-------------|-------------|-------------|------------------------|-------------------------|

|   |      |     |     |     |     |     |     |     |     |         |         |     |      |     |      |     |
|---|------|-----|-----|-----|-----|-----|-----|-----|-----|---------|---------|-----|------|-----|------|-----|
|   | From | To  |     |     |     |     |     |     |     | exposed | radical | %)  | From | To  | From | To  |
| 1 | Nil  | Nil | Nil | Nil | Nil | Nil | Nil | Nil | Nil | Nil     | Nil     | Nil | Nil  | Nil | Nil  | Nil |

2A.2.1.6: Trenching

|                    |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
|--------------------|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|
| Number of Trenches |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| Nil                |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |

2A.2.1.6.1: Spacing

|         |  |  |  |  |         |  |  |  |  |         |  |  |  |  |  |  |
|---------|--|--|--|--|---------|--|--|--|--|---------|--|--|--|--|--|--|
| Min (m) |  |  |  |  | Max (m) |  |  |  |  | Avg (m) |  |  |  |  |  |  |
| Nil     |  |  |  |  | Nil     |  |  |  |  | Nil     |  |  |  |  |  |  |

| SI.No. | Year |     | Trench ID | Length of Trench (m) | Width of Trench (m) | Depth of Trench( m) | Depth (from) | Depth(to) | Running mtr | Litho units exposed | Name of the radical | Av. Grade | Latitude (dd:mm:ss.ss) |     | Longitude (dd:mm:ss.ss) |     |
|--------|------|-----|-----------|----------------------|---------------------|---------------------|--------------|-----------|-------------|---------------------|---------------------|-----------|------------------------|-----|-------------------------|-----|
|        | From | To  |           |                      |                     |                     |              |           |             |                     |                     |           | From                   | To  | From                    | To  |
| 1      | Nil  | Nil | Nil       | Nil                  | Nil                 | Nil                 | Nil          | Nil       | Nil         | Nil                 | Nil                 | Nil       | Nil                    | Nil | Nil                     | Nil |

2A.2.1.7 Exploratory Drilling(Core/non Core)

| SI.No. | Year       |            | Exploration agency                         | Core holes                  |            | Non-core (RC/DTH)           |            | Grand total                 |            | Attach log sheet of each borehole in csv/excel format |
|--------|------------|------------|--------------------------------------------|-----------------------------|------------|-----------------------------|------------|-----------------------------|------------|-------------------------------------------------------|
|        | From       | To         |                                            | Number of boreholes drilled | Total mtrs | Number of boreholes drilled | Total mtrs | Number of boreholes drilled | Total mtrs |                                                       |
| 1      | 24/01/2020 | 12/02/2020 | Directorate of Minesand Geology, Rajasthan | 1                           | 50.00      | 0                           | 0.00       | 1                           | 50.00      | <a href="#">exploration_drilling.xlsx</a>             |

2A.2.1.8: Exploratory Mining

| SI.No. | Pit/Adit ID | Length in Mtr | Width in Mtr | Depth in mtrs | Volume (m <sup>3</sup> ) |
|--------|-------------|---------------|--------------|---------------|--------------------------|
| 1      | Nil         | Nil           | Nil          | Nil           | Nil                      |

2A.2.1.9: Sampling

| SI.No. | Type of sample | No of samples collected | Number of samples analyzed | Latitude (dd:mm:ss.ss) |             | Longitude (dd:mm:ss.ss) |             | Remark if any |
|--------|----------------|-------------------------|----------------------------|------------------------|-------------|-------------------------|-------------|---------------|
|        |                |                         |                            | From                   | To          | From                    | To          |               |
| 1      | Drill Core     | 48                      | 48                         | 26:58:14.39            | 26:58:14.39 | 73:29:23.32             | 73:29:23.32 | Nil           |

2A.2.1.10: Chemical Analysis

| SI.No. | Sample ID     | Minerals  | Radical with garde in %                                          | Name of Agency | Type of agency | Attachment                                 |
|--------|---------------|-----------|------------------------------------------------------------------|----------------|----------------|--------------------------------------------|
| 1      | TKSB-18 block | Limestone | CaO 49.81 %, SiO2 3.57 %, MgO 2.41 %, Fe2O3 0.10 %, Al2O3 0.41 % | DMG,Rajasthan  | Govt. Lab      | <a href="#">New_Chemical_analysis.xlsx</a> |

\* Chemical analysis of core /non vore samples may be uploaded in CSV file which shall normally include Five files namely collar file, survey file and Geology log file, Assay file & RQD File.

2A.2.1.11: Petrology & Mineralogical Studies

| SI.No. | Type of Sample | Number of Sample Drawn | Number of Sample Analyzed | Petrographic Study Report |
|--------|----------------|------------------------|---------------------------|---------------------------|
| 1      | Nil            | Nil                    | Nil                       | Nil                       |

2A.2.1.12: Beneficiation Studies

| SI.No. | Type of Beneficiation | Number of Samples | Attach |
|--------|-----------------------|-------------------|--------|
| 1      | Nil                   | Nil               | Nil    |

2A.2.1.13: Bulk Density Study as per M(EMC) Rules, 2015 and SOP of CGPB

Method adopted for calculating bulk density of ore and waste



The bulk density of the sample is determined byusing the formula : B.D=M/V where B.D=bulk density, M=Dry mass (weight) of the sample and V= volume of the sample

| SI.No. | Nature of Ore/OB | Mineral   | Number of samples                    | Bulk Density Established (t/m <sup>3</sup> ) |
|--------|------------------|-----------|--------------------------------------|----------------------------------------------|
| 1      | Ore              | Limestone | AspreanalysisbyDMG,Geological report | 2.50                                         |
| 2      | Overburden       | Shale     | NABL Laboratory                      | 2.50                                         |

2A.2.1.14: Area Covered under Exploration

| Level of exploration           | Area in Ha. |            | Total Area in Ha. |
|--------------------------------|-------------|------------|-------------------|
|                                | Forest      | Non Forest |                   |
| G-1                            | 0.000000    | 0.000000   | 0.000000          |
| G-2                            | 0.000000    | 4.800000   | 4.800000          |
| G-3                            | 0.000000    | 0.000000   | 0.000000          |
| G-4                            | 0.000000    | 0.000000   | 0.000000          |
| Area proved as Non-mineralized | 0.000000    | 0.000000   | 0.000000          |
| Area to be explored            | 0.000000    | 0.000000   | 0.000000          |
| Total                          | 0.000000    | 4.800000   | 4.800000          |

2A.2.2: Summary of The Previous Exploration (Before Last Plan Period)

|                    |
|--------------------|
| Name of The Agency |
| 0                  |

2A.2.2.1: Geological Mapping

| SI.No. | Year |     | Scale | Area Covered (Ha) |
|--------|------|-----|-------|-------------------|
|        | From | To  |       |                   |
| 1      | Nil  | Nil | Nil   | Nil               |

2A.2.2.2: Airborne Geophysical Survey

| SI.No. | Type of Survey | Spacing (m) | Total line (km) | Area Covered (Ha) | Latitude (dd:mm:ss.ss) |     | Longitude (dd:mm:ss.ss) |     |
|--------|----------------|-------------|-----------------|-------------------|------------------------|-----|-------------------------|-----|
|        |                |             |                 |                   | From                   | To  | From                    | To  |
| 1      | Nil            | Nil         | Nil             | Nil               | Nil                    | Nil | Nil                     | Nil |

2A.2.2.3: Ground Geophysical Survey

| SI.No. | Type of Survey | Spacing (m) | Total line (km) | Area Covered (Ha) | Latitude (dd:mm:ss.ss) |     | Longitude (dd:mm:ss.ss) |     |
|--------|----------------|-------------|-----------------|-------------------|------------------------|-----|-------------------------|-----|
|        |                |             |                 |                   | From                   | To  | From                    | To  |
| 1      | Nil            | Nil         | Nil             | Nil               | Nil                    | Nil | Nil                     | Nil |

2A.2.2.4: Geochemical Survey

| SI.No. | Type of Sample | No of Samples |
|--------|----------------|---------------|
| 1      | Nil            | Nil           |

2A.2.2.5: Pitting

| SI.No. | Pit ID | Length of Pit (m) | Width of Pit (m) | Depth of Pit (m) | Litho units exposed | Litho Unit From (m) | Litho Unit To (m) | Average Grade(%) | Running Metres (m) | Latitude (dd:mm:ss.ss) |     | Longitude (dd:mm:ss.ss) |     |
|--------|--------|-------------------|------------------|------------------|---------------------|---------------------|-------------------|------------------|--------------------|------------------------|-----|-------------------------|-----|
|        |        |                   |                  |                  |                     |                     |                   |                  |                    | Form                   | To  | Form                    | To  |
| 1      | Nil    | Nil               | Nil              | Nil              | Nil                 | Nil                 | Nil               | Nil              | Nil                | Nil                    | Nil | Nil                     | Nil |

2A.2.2.6: Trenching

|                    |
|--------------------|
| Number of Trenches |
| Nil                |



Spacing

|         |         |         |
|---------|---------|---------|
| Min (m) | Max (m) | Avg (m) |
| Nil     | Nil     | Nil     |

Area Covered Under Trenching

Co-ordinates

Latitude

|       |     |
|-------|-----|
| North | Nil |
| North | Nil |
| North | Nil |
| North | Nil |

Longitude

|      |     |
|------|-----|
| East | Nil |
| East | Nil |
| East | Nil |
| East | Nil |

| Sl.No. | Trench ID | Length of Trench (m) | Width of Trench (m) | Depth of Trench (m) | Litho Units Exposed | Average Grade | Running mtr | Latitude (dd:mm:ss.ss) |     | Longitude (dd:mm:ss.ss) |     |
|--------|-----------|----------------------|---------------------|---------------------|---------------------|---------------|-------------|------------------------|-----|-------------------------|-----|
|        |           |                      |                     |                     |                     |               |             | From                   | To  | From                    | To  |
| 1      | Nil       | Nil                  | Nil                 | Nil                 | Nil                 | Nil           | Nil         | Nil                    | Nil | Nil                     | Nil |

2A.2.2.7: Exploratory Drilling

2A.2.2.7.1:Core/Non-core Drilling

| SI.No. | Year |     | Exploration agency | Core holes                  |            | Non-core (RC/DTH)           |            | Grand total     |            | Attach log sheet of each borehole in csv/excel format |
|--------|------|-----|--------------------|-----------------------------|------------|-----------------------------|------------|-----------------|------------|-------------------------------------------------------|
|        | From | To  |                    | Number of boreholes drilled | Total mtrs | Number of boreholes drilled | Total mtrs | Total boreholes | Total mtrs |                                                       |
| 1      | Nil  | Nil | Nil                | Nil                         | Nil        | Nil                         | Nil        | Nil             | Nil        | Nil                                                   |

2A.2.2.8: Exploratory Mining

| SI.No. | Pit / Adit ID | Volume (m³) |
|--------|---------------|-------------|
| 1      | Nil           | Nil         |

2A.2.2.9: Sampling

| SI.No. | Type of sample | Number of Samples | Area Covered (Ha) | Latitude (dd:mm:ss.ss) |     | Longitude (dd:mm:ss.ss) |     |
|--------|----------------|-------------------|-------------------|------------------------|-----|-------------------------|-----|
|        |                |                   |                   | From                   | To  | From                    | To  |
| 1      | Nil            | Nil               | Nil               | Nil                    | Nil | Nil                     | Nil |

2A.2.2.10: Chemical Analysis

| SI.No. | Sample ID | Minerals | Radical Analysis | Attachment |
|--------|-----------|----------|------------------|------------|
| 1      | Nil       | Nil      | Nil              | Nil        |

2A.2.2.11:Petrology & Mineralogical Studies

| SI.No. | Type of Sample | Number of Sample Drawn | Number of Sample Analyzed | Petrographic Study Report |
|--------|----------------|------------------------|---------------------------|---------------------------|
| 1      | Nil            | Nil                    | Nil                       | Nil                       |

2A.2.2.12: Beneficiation Test

| SI.No. | Type of Beneficiation | Number of Samples | Attachment |
|--------|-----------------------|-------------------|------------|
|--------|-----------------------|-------------------|------------|

|   |     |     |     |
|---|-----|-----|-----|
| 1 | Nil | Nil | Nil |
|---|-----|-----|-----|

2A.2.2.13: Bulk Density

| SI.No. | Rock Type | Number of Samples | Minerals | Bulk Density Established (t/m <sup>3</sup> ) |
|--------|-----------|-------------------|----------|----------------------------------------------|
| 1      | Nil       | Nil               | Nil      | Nil                                          |

2A.2.2.14: Area Covered under Exploration

| Level of exploration           | Area in Ha. |            | Total Area in Ha. |
|--------------------------------|-------------|------------|-------------------|
|                                | Forest      | Non Forest |                   |
| G-1                            | 0.0000      | 0.0000     | 0.0000            |
| G-2                            | 0.0000      | 0.0000     | 0.0000            |
| G-3                            | 0.0000      | 0.0000     | 0.0000            |
| G-4                            | 0.0000      | 0.0000     | 0.0000            |
| Area proved as Non-mineralized | 0.0000      | 0.0000     | 0.0000            |
| Area to be explored            | 0.0000      | 0.0000     | 0.0000            |
| Total                          | 0.0000      | 0.0000     | 0.0000            |

| SI.No.                            | Year |     | Area converted to G1 from G2, G3 & G4 | % increase in G-1 Area | Remaining Area % in G2 | Remaining Area % in G3 | Remaining Area % in G4 | Remaining Area in G2 | Remaining Area in G3 | Remaining Area in G4 |
|-----------------------------------|------|-----|---------------------------------------|------------------------|------------------------|------------------------|------------------------|----------------------|----------------------|----------------------|
|                                   | From | To  |                                       |                        |                        |                        |                        |                      |                      |                      |
| 1                                 | Nil  | Nil | 0.00                                  | 0.00                   | 0.00                   | 0.00                   | 0.00                   | 0.00                 | 0.00                 | 0.00                 |
| Potentially Mineralised area (Ha) |      |     |                                       |                        |                        |                        |                        |                      |                      | 0.00                 |

2A.2.3 Ore Body Geometry & Grade

| SI.No. | Name of the ore band | General Strike / Trend | Dip Of Mineral Body | Average Strike Length (m) | Average Width (m) | Chemical parameters |                     |               |               |               |
|--------|----------------------|------------------------|---------------------|---------------------------|-------------------|---------------------|---------------------|---------------|---------------|---------------|
|        |                      |                        |                     |                           |                   | Average Depth (m)   | Name of the radical | Min Grade (%) | Max Grade (%) | Avg Grade (%) |

|   |           |       |   |        |        |       |     |       |       |       |
|---|-----------|-------|---|--------|--------|-------|-----|-------|-------|-------|
| 1 | Limestone | NW-SE | S | 200.00 | 240.00 | 33.34 | CaO | 47.87 | 51.67 | 49.81 |
| 2 | Limestone | NW-SE | S | 200.00 | 240.00 | 33.34 | MgO | 0.47  | 2.61  | 2.41  |

2A.2.4: Reserve / Resource Estimation Method

2A.2.4.1: Methodology

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Resource / Reserve Estimation Method                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Sectional Area Method                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Methodology                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Systematic drilling (1 nos) on grid pattern at 400m spacing aided by geological mapping has enabled in identifying Four zones of fine to medium grained grey colour limestone with a cumulative thickness is 12.50 mts in Tad as block. A total of 1 drill hole influence was considered while estimating the resources on the basis of sectional area method of which all the 1 drill hole falls within the block. The borehole spacing of 400m at this stage of investigation is quite adequate for arriving at the average grade, thickness and resources with required level of accuracy The estimations were done using Cross sectional area method in which we construct a Sections around each hole to determine an area of influence for that hole; and then the total volume directly beneath the Section is assigned the same values as the drill hole from which we constructed the Cross section. The thickness of each zone of limestone in all boreholes was calculated as per the average cut-off grade of CaO> 34%, MgO< 5% and SiO2 <16% based on geometrical mean. The area of each borehole Section is multiplied by cumulative thickness of intersected in each borehole to find volume of each borehole. The volume of limestone multiplied by specific gravity gives the resources corresponding to each zone of limestone. The specific gravity of 2.50 for limestone has been considered. The tonnages for various limestone zones of each bore hole were added to have total geological resources of limestone in block. The number of limestone zones are 4. The tonnages for various limestone zones of each bore hole were added to have total geological resources of limestone in block. |

2A.2.4.2: Resource Calculation

| SI.No. | Cross Section/Block | Section Area/ Block Area(sq mt) | Influence(m) | Depth in mtr | Volume (m³) | Bulk Density (t/m³) | Resource Quantity (t) | Level of Exploration | Type of Land | Name of the radical | Grade (%) | Method used for resource estimation |
|--------|---------------------|---------------------------------|--------------|--------------|-------------|---------------------|-----------------------|----------------------|--------------|---------------------|-----------|-------------------------------------|
| 1      | TSB61               | 48000                           | 0.00         | 12.50        | 600000.00   | 2.50                | 1500000.00000         | 332                  | Govt. & Pvt. | CaO                 | 49.81     | Polygonarea method                  |
| Total  |                     |                                 |              |              | 600000.00   |                     | 1500000.00000         |                      |              |                     |           |                                     |

2A.2.4.3: Mineral Resource Estimate for Conversion to Mineral Reserve

All the estimation has been done as per Part III of The Minerals (Evidence of Mineral Contents) Rules, 2015. The deposits of regular habit & prospected at 400m×400m grid pattern. Depth of the bore holes 50 m. The limestone deposit of the area is of regular habit and it fall under G-2 category, as per Part III of The Minerals (Evidence of Mineral Contents) Rules, 2015. In the block area of 4.80 ha 1 core bore holes were considered as per Geological report provided along with tender document by DMG Rajasthan.

**2A.2.4.4: Threshold value & Cut off Parameters**

There hold Value of Limestone mineral is  $\text{CaO} > 34\%$ ,  $\text{MgO} < 5\%$  and  $\text{SiO}_2 < 16\%$  (max) as per IBM notification dated 25/04/2018. The cut off grade of limestone mineral for this block area has also been considered same as per their hold value of limestone mineral. The thickness of each zone of limestone in all boreholes was calculated as per the average cut-off grade of  $\text{CaO} > 34\%$ ,  $\text{MgO} < 5\%$  and  $\text{SiO}_2 < 16\%$  based on geometrical mean.

**2A.2.4.5: Mining Factors or Assumptions**

Following mining factor or assumption have been considered: \*Buffer zone of 7.5m width all along the periphery of concession area. The number of mineral zone is from 1 to 5. The thickness of limestone considered on the basis of exploratory drilling bore holes data i.e. from 5.10m (Min.) to 38.00m (Max) and average strike length and width of limestone band in block area are 200 mtr and 240 mtrs respectively. \* Minimum safety distance of 10m on either side of Cart road passing through the block area. \*Bulk Density of limestone assumed as 2.5 t/m<sup>3</sup>. \*Minimum Bench width 6m in mineral zone & 6m in OB. \*Bench height of 6m for mineral & OB \*Ultimate pit slope is 37 degree.

**2A.2.4.6: Metallurgical Factors or Assumptions**

The limestone produced in this mine will feed directly to Lime kiln or sell in the market, thus the metallurgical factor does not apply.

**2A.2.4.7: Cost & Revenue Factors**

The chemical grade limestone cost is Rs-501 per tonne with it includes Mining cost, Royalty, DMF, NMET, Auction premium, Cess on royalty etc.

**2A.2.4.8: Market Assessment**

Total quantity of limestone production will Sell to Lime kiln for manufacturing Hydrated lime in Jodhpur and Nagaur District.

**2A.2.4.9: Other Modifying Factors**

Currently, there are no national parks or historical monuments near the block area, and no tribal issues are anticipated. The auction block area of 4.80 ha consists of government land and private land. No mining activities will take place before the grant of a Mining Lease. Suncity Chemical & Minerals Private Limited will fulfill all statutory and legal obligations and special conditions imposed, if any, by the statutory agencies: a) During the grant and execution of the lease deed by the Directorate of Mines and Geology, Government of Rajasthan, b) Directives issued by the Indian Bureau of Mines, c) Directives issued by the Directorate General of Mines Safety, d) Conditions imposed by the Ministry of Environment and Forests and Climate Change, as well as the State or Central Pollution Control Board, and any other statutory organization outlining the nature of conditions and compliance status.

#### 2A.2.4.10: Classification

The deposit type is a bedded stratiform and tabular deposit of regular habit as per the MEMC Rules 2015. The block has a total resource of 1.50 million tonnes. The probable mineral reserve under category 122 is 0.978004 million tonnes, and the pre-feasibility mineral resource under category 222 is 0.521996 million tonnes.

#### 2A.2.4.11: Calculation of blocked resources

| Sl.No. | Reserves blocked due to  | Cross section/Block | Sectional area/ block area (in Sq mtr) | Influence (m) | Depth (m) | Volume (m <sup>3</sup> ) | Bulk Density (t/m <sup>3</sup> ) | Resource Quantity (t) | UNFC code | Type of Land | Name of the radical | Grade (%)  | Method used for resource estimation |
|--------|--------------------------|---------------------|----------------------------------------|---------------|-----------|--------------------------|----------------------------------|-----------------------|-----------|--------------|---------------------|------------|-------------------------------------|
| 1      | 7.5 Meter Safety Barrier | Layer 1 QQ          | 27.97                                  | 0.00          | 45.00     | 1258.65                  | 2.50                             | 3146.62500            | 222       | Govt.& Pvt.  | CaOMgO              | 49.81,2.41 | Cross Section                       |
| 2      | 7.5 Meter Safety Barrier | Layer 1 RR          | 27.97                                  | 0.00          | 35.00     | 978.95                   | 2.50                             | 2447.37500            | 222       | Govt.& Pvt.  | CaOMgO              | 49.81,2.41 | Cross Section                       |
| 3      | 7.5 Meter Safety Barrier | Layer 1 SS          | 27.97                                  | 0.00          | 35.00     | 978.95                   | 2.50                             | 2447.37500            | 222       | Govt.& Pvt.  | CaOMgO              | 49.81,2.41 | Cross Section                       |
| 4      | 7.5 Meter Safety Barrier | Layer 1 TT          | 27.97                                  | 0.00          | 33.00     | 923.01                   | 2.50                             | 2307.52500            | 222       | Govt.& Pvt.  | CaOMgO              | 49.81,2.41 | Cross Section                       |
| 5      | 7.5 Meter Safety Barrier | Layer 1 UU          | 27.94                                  | 0.00          | 35.00     | 977.90                   | 2.50                             | 2444.75000            | 222       | Govt.& Pvt.  | CaOMgO              | 49.81,2.41 | Cross Section                       |
| 6      | 7.5 Meter                | Layer 2 QQ          | 75.00                                  | 0.00          | 45.00     | 3375.00                  | 2.50                             | 8437.50000            | 222       | Govt.& Pvt.  | CaOMgO              | 49.81,2.41 | Cross                               |



|    | Safety Barrier           |            |        |      |       |          |      |             |     |              |        |            | Section       |
|----|--------------------------|------------|--------|------|-------|----------|------|-------------|-----|--------------|--------|------------|---------------|
| 7  | 7.5 Meter Safety Barrier | Layer 2 RR | 75.00  | 0.00 | 35.00 | 2625.00  | 2.50 | 6562.50000  | 222 | Govt.& Pvt.  | CaOMgO | 49.81,2.41 | Cross Section |
| 8  | 7.5 Meter Safety Barrier | Layer 2 SS | 75.00  | 0.00 | 35.00 | 2625.00  | 2.50 | 6562.50000  | 222 | Govt.& Pvt.  | CaOMgO | 49.81,2.41 | Cross Section |
| 9  | 7.5 Meter Safety Barrier | Layer 2 TT | 75.00  | 0.00 | 33.00 | 2475.00  | 2.50 | 6187.50000  | 222 | Govt.& Pvt.  | CaOMgO | 49.81,2.41 | Cross Section |
| 10 | 7.5 Meter Safety Barrier | Layer 2 UU | 75.00  | 0.00 | 35.00 | 2625.00  | 2.50 | 6562.50000  | 222 | Govt.& Pvt.  | CaOMgO | 49.81,2.41 | Cross Section |
| 11 | 7.5 Meter Safety Barrier | Layer 3 QQ | 382.00 | 0.00 | 45.00 | 17190.00 | 2.50 | 42975.00000 | 222 | Govt.& Pvt.  | CaOMgO | 49.81,2.41 | Cross Section |
| 12 | 7.5 Meter Safety Barrier | Layer 3 RR | 382.00 | 0.00 | 35.00 | 13370.00 | 2.50 | 33425.00000 | 222 | Govt.& Pvt.  | CaOMgO | 49.81,2.41 | Cross Section |
| 13 | 7.5 Meter Safety Barrier | Layer 3 SS | 382.00 | 0.00 | 35.00 | 13370.00 | 2.50 | 33425.00000 | 222 | Govt.& Pvt.  | CaOMgO | 49.81,2.41 | Cross Section |
| 14 | 7.5 Meter Safety Barrier | Layer 3 TT | 382.00 | 0.00 | 33.00 | 12606.00 | 2.50 | 31515.00000 | 222 | Govt.& Pvt.  | CaOMgO | 49.81,2.41 | Cross Section |
| 15 | 7.5 Meter Safety Barrier | Layer 3 UU | 382.00 | 0.00 | 35.00 | 13370.00 | 2.50 | 33425.00000 | 222 | Govt.& Pvt.  | CaOMgO | 49.81,2.41 | Cross Section |
| 16 | 7.5 Meter Safety Barrier | Layer 4 QQ | 448.00 | 0.00 | 45.00 | 20160.00 | 2.50 | 50400.00000 | 222 | Govt.& Pvt.  | CaOMgO | 49.81,2.41 | Cross Section |
| 17 | 7.5 Meter Safety Barrier | Layer 4 RR | 448.00 | 0.00 | 35.00 | 15680.00 | 2.50 | 39200.00000 | 222 | Govt. & Pvt. | CaOMgO | 49.81,2.41 | Cross Section |

|    |                          |            |        |      |       |          |      |             |     |              |        |            |               |
|----|--------------------------|------------|--------|------|-------|----------|------|-------------|-----|--------------|--------|------------|---------------|
| 18 | 7.5 Meter Safety Barrier | Layer 4 SS | 448.00 | 0.00 | 35.00 | 15680.00 | 2.50 | 39200.00000 | 222 | Govt. & Pvt. | CaOMgO | 49.81,2.41 | Cross Section |
| 19 | 7.5 Meter Safety Barrier | Layer 4 TT | 448.00 | 0.00 | 33.00 | 14784.00 | 2.50 | 36960.00000 | 222 | Govt. & Pvt. | CaOMgO | 49.81,2.41 | Cross Section |
| 20 | 7.5 Meter Safety Barrier | Layer 4 UU | 448.00 | 0.00 | 37.00 | 16576.00 | 2.50 | 41440.00000 | 222 | Govt. & Pvt. | CaOMgO | 49.81,2.41 | Cross Section |
| 21 | Ultimate Pit Limit       | Layer 1 QQ | 22.50  | 0.00 | 52.50 | 1181.25  | 2.50 | 2953.12500  | 222 | Govt. & Pvt. | CaOMgO | 49.81,2.41 | Cross Section |
| 22 | Ultimate Pit Limit       | Layer 1 RR | 22.50  | 0.00 | 35.00 | 787.50   | 2.50 | 1968.75000  | 222 | Govt. & Pvt. | CaOMgO | 49.81,2.41 | Cross Section |
| 23 | Ultimate Pit Limit       | Layer 1 SS | 22.50  | 0.00 | 35.00 | 787.50   | 2.50 | 1968.75000  | 222 | Govt. & Pvt. | CaOMgO | 49.81,2.41 | Cross Section |
| 24 | Ultimate Pit Limit       | Layer 1 TT | 22.50  | 0.00 | 33.00 | 742.50   | 2.50 | 1856.25000  | 222 | Govt. & Pvt. | CaOMgO | 49.81,2.41 | Cross Section |
| 25 | Ultimate Pit Limit       | Layer 1 UU | 22.50  | 0.00 | 44.50 | 1001.25  | 2.50 | 2503.12500  | 222 | Govt. & Pvt. | CaOMgO | 49.81,2.41 | Cross Section |
| 26 | Ultimate Pit Limit       | Layer 2 QQ | 15.00  | 0.00 | 52.50 | 787.50   | 2.50 | 1968.75000  | 222 | Govt. & Pvt. | CaOMgO | 49.81,2.41 | Cross Section |
| 27 | Ultimate Pit Limit       | Layer 2 RR | 15.00  | 0.00 | 35.00 | 525.00   | 2.50 | 1312.50000  | 222 | Govt. & Pvt. | CaOMgO | 49.81,2.41 | Cross Section |
| 28 | Ultimate Pit Limit       | Layer 2 SS | 15.00  | 0.00 | 35.00 | 525.00   | 2.50 | 1312.50000  | 222 | Govt. & Pvt. | CaOMgO | 49.81,2.41 | Cross Section |
| 29 | Ultimate Pit Limit       | Layer 2 TT | 15.00  | 0.00 | 33.00 | 495.00   | 2.50 | 1237.50000  | 222 | Govt. & Pvt. | CaOMgO | 49.81,2.41 | Cross Section |
| 30 | Ultimate Pit Limit       | Layer 2 UU | 15.00  | 0.00 | 42.50 | 637.50   | 2.50 | 1593.75000  | 222 | Govt. & Pvt. | CaOMgO | 49.81,2.41 | Cross Section |
| 31 | Ultimate Pit Limit       | Layer 3 QQ | 75.00  | 0.00 | 52.50 | 3937.50  | 2.50 | 9843.75000  | 222 | Govt. & Pvt. | CaOMgO | 49.81,2.41 | Cross Section |
| 32 | Ultimate Pit Limit       | Layer 3 RR | 75.00  | 0.00 | 35.00 | 2625.00  | 2.50 | 6562.50000  | 222 | Govt. & Pvt. | CaOMgO | 49.81,2.41 | Cross Section |



|       |                    |            |       |      |       |           |      |            |     |              |        |            |               |
|-------|--------------------|------------|-------|------|-------|-----------|------|------------|-----|--------------|--------|------------|---------------|
| 33    | Ultimate Pit Limit | Layer 3 SS | 75.00 | 0.00 | 35.00 | 2625.00   | 2.50 | 6562.50000 | 222 | Govt. & Pvt. | CaOMgO | 49.81,2.41 | Cross Section |
| 34    | Ultimate Pit Limit | Layer 3 TT | 75.00 | 0.00 | 33.00 | 2475.00   | 2.50 | 6187.50000 | 222 | Govt. & Pvt. | CaOMgO | 49.81,2.41 | Cross Section |
| 35    | Ultimate Pit Limit | Layer 3 UU | 75.00 | 0.00 | 42.50 | 3187.50   | 2.50 | 7968.75000 | 222 | Govt. & Pvt. | CaOMgO | 49.81,2.41 | Cross Section |
| 36    | Ultimate Pit Limit | Layer 3 QQ | 75.00 | 0.00 | 52.50 | 3937.50   | 2.50 | 9843.75000 | 222 | Govt. & Pvt. | CaOMgO | 49.81,2.41 | Cross Section |
| 37    | Ultimate Pit Limit | Layer 3 RR | 75.00 | 0.00 | 35.00 | 2625.00   | 2.50 | 6562.50000 | 222 | Govt. & Pvt. | CaOMgO | 49.81,2.41 | Cross Section |
| 38    | Ultimate Pit Limit | Layer 3 SS | 75.00 | 0.00 | 35.00 | 2625.00   | 2.50 | 6562.50000 | 222 | Govt. & Pvt. | CaOMgO | 49.81,2.41 | Cross Section |
| 39    | Ultimate Pit Limit | Layer 3 TT | 75.00 | 0.00 | 33.00 | 2475.00   | 2.50 | 6187.50000 | 222 | Govt. & Pvt. | CaOMgO | 49.81,2.41 | Cross Section |
| 40    | Ultimate Pit Limit | Layer 3 UU | 75.00 | 0.00 | 42.50 | 3187.50   | 2.50 | 7968.75000 | 222 | Govt. & Pvt. | CaOMgO | 49.81,2.41 | Cross Section |
| Total |                    |            |       |      |       | 208798.46 |      | 521996.15  |     |              |        |            |               |

2A.2.4.12: Calulation of Reserves - I

| SI.No. | Cross section/Bloc k | Sectional area/ block area (in Sq mtr) | Influence (m) | Depth (m) | Volume (m³) | Bulk Density (t/m³) | Resource Quantity (t) | UNFC code | Type of Land | Name of the radical | Grade (%)  | Method used for resource estimation |
|--------|----------------------|----------------------------------------|---------------|-----------|-------------|---------------------|-----------------------|-----------|--------------|---------------------|------------|-------------------------------------|
| 1      | Layer 1 QQ           | 370                                    | 0.00          | 45.00     | 16650.00    | 2.50                | 41625.00              | 122       | Govt.& Pvt   | CaOMgO              | 49.81,2.41 | Cross Section                       |
| 2      | Layer 1 RR           | 370                                    | 0.00          | 35.00     | 12950.00    | 2.50                | 32375.00              | 122       | Govt.& Pvt   | CaOMgO              | 49.81,2.41 | Cross Section                       |
| 3      | Layer 1 SS           | 370                                    | 0.00          | 35.00     | 12950.00    | 2.50                | 32375.00              | 122       | Govt.& Pvt   | CaOMgO              | 49.81,2.41 | Cross Section                       |
| 4      | Layer 1 TT           | 370                                    | 0.00          | 33.00     | 12210.00    | 2.50                | 30525.00              | 122       | Govt.& Pvt   | CaOMgO              | 49.81,2.41 | Cross Section                       |

|       |            |       |      |       |           |      |           |     |            |        |            |               |
|-------|------------|-------|------|-------|-----------|------|-----------|-----|------------|--------|------------|---------------|
| 5     | Layer 1 UU | 370   | 0.00 | 35.00 | 12950.00  | 2.50 | 32375.00  | 122 | Govt.& Pvt | CaOMgO | 49.81,2.41 | Cross Section |
| 6     | Layer 2 QQ | 370.1 | 0.00 | 45.00 | 16654.50  | 2.50 | 41636.25  | 122 | Govt.& Pvt | CaOMgO | 49.81,2.41 | Cross Section |
| 7     | Layer 2 RR | 370.1 | 0.00 | 35.00 | 12953.50  | 2.50 | 32383.75  | 122 | Govt.& Pvt | CaOMgO | 49.81,2.41 | Cross Section |
| 8     | Layer 2 SS | 370.1 | 0.00 | 35.00 | 12953.50  | 2.50 | 32383.75  | 122 | Govt.& Pvt | CaOMgO | 49.81,2.41 | Cross Section |
| 9     | Layer 2 TT | 370.1 | 0.00 | 33.00 | 12213.30  | 2.50 | 30533.25  | 122 | Govt.& Pvt | CaOMgO | 49.81,2.41 | Cross Section |
| 10    | Layer 2 UU | 370.1 | 0.00 | 35.00 | 12953.50  | 2.50 | 32383.75  | 122 | Govt.& Pvt | CaOMgO | 49.81,2.41 | Cross Section |
| 11    | Layer 3 QQ | 795.1 | 0.00 | 45.00 | 35779.50  | 2.50 | 89448.75  | 122 | Govt.& Pvt | CaOMgO | 49.81,2.41 | Cross Section |
| 12    | Layer 3 RR | 795.1 | 0.00 | 35.00 | 27828.50  | 2.50 | 69571.25  | 122 | Govt.& Pvt | CaOMgO | 49.81,2.41 | Cross Section |
| 13    | Layer 3 SS | 795.1 | 0.00 | 35.00 | 27828.50  | 2.50 | 69571.25  | 122 | Govt.& Pvt | CaOMgO | 49.81,2.41 | Cross Section |
| 14    | Layer 3 TT | 795.1 | 0.00 | 33.00 | 26238.30  | 2.50 | 65595.75  | 122 | Govt.& Pvt | CaOMgO | 49.81,2.41 | Cross Section |
| 15    | Layer 3 UU | 795.1 | 0.00 | 35.00 | 27828.50  | 2.50 | 69571.25  | 122 | Govt.& Pvt | CaOMgO | 49.81,2.41 | Cross Section |
| 16    | Layer 4 QQ | 745   | 0.00 | 45.00 | 33525.00  | 2.50 | 83812.50  | 122 | Govt.& Pvt | CaOMgO | 49.81,2.41 | Cross Section |
| 17    | Layer 4 RR | 745   | 0.00 | 35.00 | 26075.00  | 2.50 | 65187.50  | 122 | Govt.& Pvt | CaOMgO | 49.81,2.41 | Cross Section |
| 18    | Layer 4 SS | 745   | 0.00 | 35.00 | 26075.00  | 2.50 | 65187.50  | 122 | Govt.& Pvt | CaOMgO | 49.81,2.41 | Cross Section |
| 19    | Layer 4 TT | 745   | 0.00 | 33.00 | 24585.00  | 2.50 | 61462.50  | 122 | Govt.& Pvt | CaOMgO | 49.81,2.41 | Cross Section |
| Total |            |       |      |       | 391201.60 |      | 978004.00 |     |            |        |            |               |

2A.2.4.13: Calculation of Reserves -II

|                                     |            |
|-------------------------------------|------------|
| Mineral                             | LIMESTONE  |
| Reserves/ Resources estimated as on | 10/03/2025 |
| UNIT of estimation                  | tonnes     |

A. Mineral Reserve

| Classification                  | Code | Quantity |            |           | Grade  |                      | Remark |
|---------------------------------|------|----------|------------|-----------|--------|----------------------|--------|
|                                 |      | Forest   | Non Forest | Total     | Forest | Non Forest           |        |
| 1. Proved Mineral Reserve (A)   | 111  | 0.00     | 0.00       | 0.00      | 0.00   | 0.00                 | Nil    |
| 2. Probable Mineral Reserve (A) | 121  | 0.00     | 0.00       | 0.00      | 0.00   | 0.00                 | Nil    |
| 3. Probable Mineral Reserve (A) | 122  | 0.00     | 978004.00  | 978004.00 | Nil    | Ca-49.81, MgO - 2.41 | Nil    |

B. Remaining Resources

| Classification                         | Code | Quantity |            |           | Grade  |                      | Remark |
|----------------------------------------|------|----------|------------|-----------|--------|----------------------|--------|
|                                        |      | Forest   | Non Forest | Total     | Forest | Non Forest           |        |
| 1. Feasibility Mineral Resource (B)    | 211  | 0.00     | 0.00       | 0.00      | 0      | 0                    | Nil    |
| 2. Prefeasibility Mineral Resource (B) | 221  | 0.00     | 0.00       | 0.00      | 0      | 0                    | Nil    |
| 3. Prefeasibility Mineral Resource (B) | 222  | 0.00     | 521996.00  | 521996.00 | 0      | Ca-49.81, MgO - 2.41 | Nil    |
| 4. Measured Mineral Resource (B)       | 331  | 0.00     | 0.00       | 0.00      | 0      | 0                    | Nil    |
| 5. Indicated Mineral Resource (B)      | 332  | 0.00     | 0.00       | 0.00      | 0      | 0                    | Nil    |

|                                        |     |      |      |            |   |   |     |
|----------------------------------------|-----|------|------|------------|---|---|-----|
| 6. Inferred Mineral Resource (B)       | 333 | 0.00 | 0.00 | 0.00       | 0 | 0 | Nil |
| 7. Reconnaissance Mineral Resource (B) | 334 | 0.00 | 0.00 | 0.00       | 0 | 0 | Nil |
| Total Mineral Resources (A+B) :        |     |      |      | 1500000.00 |   |   |     |

2A.2.4.13: Calculation of Reserves -III

No associate minerals are available!

2A.2.5: Future Exploration Proposal

2A.2.5.1: Geological Mapping

|       |       |        |                   |
|-------|-------|--------|-------------------|
| SI.N. | Year  | Scale  | Area Covered (Ha) |
| 1     | Year1 | 1:1000 | 4.80              |

2A.2.5.2: Ground Geophysical Survey

| SI.No. | Year | Type of Survey | Spacing (m) | Total line (km) | Area Covered (Ha) | Latitude (dd:mm:ss.ss) |     | Longitude (dd:mm:ss.ss) |     |
|--------|------|----------------|-------------|-----------------|-------------------|------------------------|-----|-------------------------|-----|
|        |      |                |             |                 |                   | From                   | To  | From                    | To  |
| 1      | Nil  | Nil            | Nil         | Nil             | Nil               | Nil                    | Nil | Nil                     | Nil |

2A.2.5.3: Pitting

|                |      |           |        |                      |                     |                     |                        |    |                         |    |
|----------------|------|-----------|--------|----------------------|---------------------|---------------------|------------------------|----|-------------------------|----|
| Number of Pits |      |           |        |                      |                     |                     |                        |    |                         |    |
| Nil            |      |           |        |                      |                     |                     |                        |    |                         |    |
| SI.No.         | Year | Land Type | Pit ID | Length of Pit<br>(m) | Width of Pit<br>(m) | Depth of Pit<br>(m) | Latitude (dd:mm:ss.ss) |    | Longitude (dd:mm:ss.ss) |    |
|                |      |           |        |                      |                     |                     | From                   | To | From                    | To |

|   |     |     |     |     |     |     |     |     |     |     |
|---|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| 1 | Nil | Nil | Nil | Nil | Nil | Nil | Nil | Nil | Nil | Nil |
|---|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|

2A.2.5.4: Trenching

|                    |
|--------------------|
| Number of Trenches |
| Nil                |

2A.2.5.4.1: SPACING

|         |         |         |
|---------|---------|---------|
| Min (m) | Max (m) | Avg (m) |
| Nil     | Nil     | Nil     |

2A.2.5.4.2: Area Covered Under Trenching

Co-ordinates

| Sl.No. | Year | Land Type | Trench ID | Length of Trench (m) | Width of Trench (m) | Depth of Trench(m) | Latitude (dd:mm:ss.ss) |     | Longitude (dd:mm:ss.ss) |     |
|--------|------|-----------|-----------|----------------------|---------------------|--------------------|------------------------|-----|-------------------------|-----|
|        |      |           |           |                      |                     |                    | From                   | To  | From                    | To  |
| 1      | Nil  | Nil       | Nil       | Nil                  | Nil                 | Nil                | Nil                    | Nil | Nil                     | Nil |

2A.2.5.5: Exploratory Drilling

2A.2.5.5.1: Core Drilling & Non-Core Drilling

| Sl.No. | Year  | In Forest Area  |           |               |               | In Non Forest Area |           |               |               | Total Borehole | Total Meter |
|--------|-------|-----------------|-----------|---------------|---------------|--------------------|-----------|---------------|---------------|----------------|-------------|
|        |       | No. of Borehole | Total Mtr | Type Borehole | Grid Interval | No. of Borehole    | Total Mtr | Type Borehole | Grid Interval |                |             |
| 1      | Year1 | 0               | 0.00      | Nil           | 0.00          | 3                  | 180.00    | Core          | 100.00        | 3              | 180.00      |

2A.2.5.6: Exploratory Mining

| SI.No. | Year | Pit ID | Length in meter | Width in meter | Depth in meter | Volume (m³) |
|--------|------|--------|-----------------|----------------|----------------|-------------|
| 1      | Nil  | Nil    | Nil             | Nil            | Nil            | Nil         |

2A.2.5.7: Sampling

| SI.No. | Year  | Type of Sample | Number of Samples Proposed | Area Covered(Ha) | Latitude (dd:mm:ss.ss) |             | Longitude (dd:mm:ss.ss) |             |
|--------|-------|----------------|----------------------------|------------------|------------------------|-------------|-------------------------|-------------|
|        |       |                |                            |                  | From                   | To          | From                    | To          |
| 1      | Year1 | Drill Core     | 180                        | 4.80             | 26:58:14.33            | 26:58:11.08 | 73:29:19.72             | 73:29:23.37 |

2A.2.5.8 Petrographic & Mineralgraphic Studies

| SI.No. | Year | Type of Sample | Number of Samples Proposed |
|--------|------|----------------|----------------------------|
| 1      | Nil  | Nil            | Nil                        |



# **Chapter 2B : Geology & Exploration UG : NA**

Approved

Chapter 3: Mineral Beneficiation / Processing

|                         |                          |
|-------------------------|--------------------------|
| Name of The Ore/Mineral | Limestone Chemical Grade |
|-------------------------|--------------------------|

3.1: Mineralogy of the ROM ore/ Mineral

|       |                        |                   |                       |                          |
|-------|------------------------|-------------------|-----------------------|--------------------------|
| SI.No | Valuable Mineral Name  | Approx. Mineral % | Gangue Mineral/s name | Approx. Mineral Gangue % |
| 1     | Limestone Cement Grade | 100.0000          | Nil                   | 0.0000                   |

3.2: Complete Chemical Analysis of the ROM Ore/Mineral

|       |         |         |
|-------|---------|---------|
| SI.No | Radical | Wt%     |
| 1     | CaO     | 49.8100 |
| 2     | SiO2    | 3.5700  |
| 3     | MgO     | 2.4100  |
| 4     | Fe2O3   | 0.1000  |
| 5     | Al2O3   | 0.4100  |

3.3: Crushing Section

3.3.1: Primary Crushing

Not Applicable

3.3.2: Secondary Crushing



Not Applicable

**3.3.3: Tertiary Crushing**

Not Applicable

**3.4: Grinding Section**

**3.4.1: Dry Grinding**

Not Applicable

**3.4.2: Wet Grinding**

Not Applicable

**3.5: Dry Processing**

**3.5.1: Screening and Classification**

Not Applicable

**3.5.2: Other Operations**

Not Applicable

3.5.3: Product Quality

Not Applicable

3.6: Wet Processing

3.6.1: Scrubbing / Washing

| SI.No | Type of Scrubbers / washers | Stages, if applicable | Make | Capacity(tph) | Feed Size(mm) | Product Size (mm) | Product Quality, if available | Water Requirement(l/h) | Fresh Water Requirement (l/h) | Recirculated Water (l/h) |
|-------|-----------------------------|-----------------------|------|---------------|---------------|-------------------|-------------------------------|------------------------|-------------------------------|--------------------------|
| 1     | Nil                         | Nil                   | Nil  | Nil           | Nil           | Nil               | Nil                           | Nil                    | Nil                           | Nil                      |

3.6.2: Screening and Classification

| SI.No | Type of screen / classifiers | Stages, if applicable | Make | Capacity(tph) | Aperture Size of Screen/Classifier (mm), if applicable | Feed Size(mm) | Product Size (mm) | Product Quality, if available | Water Requirement(l/h) | Fresh Water Requirement (l/h) | Recirculated Water (l/h) |
|-------|------------------------------|-----------------------|------|---------------|--------------------------------------------------------|---------------|-------------------|-------------------------------|------------------------|-------------------------------|--------------------------|
| 1     | Nil                          | Nil                   | Nil  | Nil           | Nil                                                    | Nil           | Nil               | Nil                           | Nil                    | Nil                           | Nil                      |

3.6.3: Gravity Separation

| SI.No | Type of separators (jig, table, | Stages, if applicable | Make | Capacity(tph) | Feed Size(mm) | Product (Conc) (tph) | Product-Mid (tph), if available | Product-Tail (tph) | Water Requirement(l/h) | Fresh Water Requirement (l/h) | Recirculated Water (l/h) |
|-------|---------------------------------|-----------------------|------|---------------|---------------|----------------------|---------------------------------|--------------------|------------------------|-------------------------------|--------------------------|
|-------|---------------------------------|-----------------------|------|---------------|---------------|----------------------|---------------------------------|--------------------|------------------------|-------------------------------|--------------------------|

|   |               |     |     |     |     |     |     |     |     |     |     |
|---|---------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
|   | spiral, etc.) |     |     |     |     |     |     |     |     |     |     |
| 1 | Nil           | Nil | Nil | Nil | Nil | Nil | Nil | Nil | Nil | Nil | Nil |

3.6.4: Magnetic Separation

| SI.No | Type of magnetic separators (magnetic intensity) | Stages, if applicable | Make | Capacity(tph) | Feed Size(mm) | Product-Mag (tph) | Product-Mid (tph), if available | Product non-Mag (tph) | Water Requirement(l/h) | Fresh Water Requirement (l/h) | Recirculated Water (l/h) |
|-------|--------------------------------------------------|-----------------------|------|---------------|---------------|-------------------|---------------------------------|-----------------------|------------------------|-------------------------------|--------------------------|
| 1     | Nil                                              | Nil                   | Nil  | Nil           | Nil           | Nil               | Nil                             | Nil                   | Nil                    | Nil                           | Nil                      |

3.6.5: Flotation

| SI.No | Type of flotation equipment (froth/ column) | Stages (rougher/ cleaner, etc), if applicable | Make | Capacity(tph) | Feed Size(mm) | Product-Float (tph) | Product non-Float (tph) | Water Requirement(l/h) | Fresh Water Requirement (l/h) | Recirculated Water (l/h) |
|-------|---------------------------------------------|-----------------------------------------------|------|---------------|---------------|---------------------|-------------------------|------------------------|-------------------------------|--------------------------|
| 1     | Nil                                         | Nil                                           | Nil  | Nil           | Nil           | Nil                 | Nil                     | Nil                    | Nil                           | Nil                      |

3.6.6: Other Operations

| SI.No | Type of equipment / operation | Stages, if applicable | Make | Capacity(tph) | Feed Size(mm) | Product-Conc (tph) | Product-Mid (tph), if available | Product-Tail (tph) | Water Requirement(l/h) | Fresh Water Requirement (l/h) | Recirculated Water (l/h) |
|-------|-------------------------------|-----------------------|------|---------------|---------------|--------------------|---------------------------------|--------------------|------------------------|-------------------------------|--------------------------|
| 1     | Nil                           | Nil                   | Nil  | Nil           | Nil           | Nil                | Nil                             | Nil                | Nil                    | Nil                           | Nil                      |

3.6.7: Product Quality (wet processing)

|          |     |           |                 |                            |
|----------|-----|-----------|-----------------|----------------------------|
| Products | Wt% | In Tonnes | Size (Range) mm | Complete chemical analysis |
|----------|-----|-----------|-----------------|----------------------------|

|             |     |     |     |     |
|-------------|-----|-----|-----|-----|
| Concentrate | Nil | Nil | Nil | Nil |
| Sub-grade   | Nil | Nil | Nil | Nil |
| Rejects     | Nil | Nil | Nil | Nil |

3.7: Overall Product Quality (Dry cum Wet Processing)

|             |     |           |                 |                            |
|-------------|-----|-----------|-----------------|----------------------------|
| Products    | Wt% | In Tonnes | Size (Range) mm | Complete chemical analysis |
| Concentrate | Nil | Nil       | Nil             | Nil                        |
| Sub-grade   | Nil | Nil       | Nil             | Nil                        |
| Rejects     | Nil | Nil       | Nil             | Nil                        |

3.8: Disposal Method for tailing/ rejects

|                                                                                                                               |     |
|-------------------------------------------------------------------------------------------------------------------------------|-----|
| a) Explain the disposal method for tailing or reject from processing plant with detail chemical / mineral analysis of tailing | Nil |
| b) Size and capacity of tailing pond, toxic effect of such tailings, process adopted to neutralise its effect (if any)        | Nil |
| c) Any other data (if available)                                                                                              | Nil |

3.9: Overall water requirement of mining and mineral processing

|                                                                                                               |                                           |
|---------------------------------------------------------------------------------------------------------------|-------------------------------------------|
| Indicate quantity, source of supply, disposal of water and extent of recycling and chemical analysis of water | <a href="#">Updated water balance.pdf</a> |
|---------------------------------------------------------------------------------------------------------------|-------------------------------------------|

3.10: Flow sheets and charts

|                                                                               |     |
|-------------------------------------------------------------------------------|-----|
| Material balance chart of mineral processing plant(s) (each stage of process) | Nil |
| Attach flow sheet of beneficiation of plant(s)                                | Nil |
|                                                                               |     |

|                                |     |
|--------------------------------|-----|
| Any other data (if applicable) | Nil |
|--------------------------------|-----|

Approved

Chapter 4A: Mining Operations

|                                   |                             |                                    |                            |      |
|-----------------------------------|-----------------------------|------------------------------------|----------------------------|------|
| 4A.1.1: Existing Method of Mining |                             |                                    | Nil                        |      |
| 4A.1.2: Proposed Method of Mining |                             |                                    | Mechanized                 |      |
| Choose one or more                | HEMM with deephole drilling | Combination of loaders and tippers | None                       | None |
| Reasons for Proposed Changes      |                             |                                    | Not Applicable fresh grant |      |

4A.2: Operational Parameters

4A.2.1: Inventory of Existing Pits & Dumps

4A.2.1.1: Pits

| SI.No. | Pit ID | Pit Status | Area Covered by Pit(Ha) | Pit Dimensions(L*W*D) |
|--------|--------|------------|-------------------------|-----------------------|
| 1      | Nil    | Nil        | Nil                     | Nil                   |

4A.2.1.2: Dumps and Stacks

4A.2.1.2.1: Dump Details

| SI.No. | Dump ID | Dump Status | Type of Dump | Total of Dump Quantity(t) | Area Covered by Dump(Ha) | Height(m) | Latitude (dd:mm:ss.ss) | Longitude (dd:mm:ss.ss) |
|--------|---------|-------------|--------------|---------------------------|--------------------------|-----------|------------------------|-------------------------|
|--------|---------|-------------|--------------|---------------------------|--------------------------|-----------|------------------------|-------------------------|



|   |     |     |     |     |     |     |      |     |      |     |
|---|-----|-----|-----|-----|-----|-----|------|-----|------|-----|
|   |     |     |     |     |     |     | From | To  | From | To  |
| 1 | Nil | Nil | Nil | Nil | Nil | Nil | Nil  | Nil | Nil  | Nil |

4A.2.1.2.2: Stack Details

| Sl.No. | Stack ID | Type of Stack | Total Stack of Quantity(t) | Area Covered by Stack(Ha) | Height(m) | Latitude (dd:mm:ss.ss) |     | Longitude (dd:mm:ss.ss) |     |
|--------|----------|---------------|----------------------------|---------------------------|-----------|------------------------|-----|-------------------------|-----|
|        |          |               |                            |                           |           | From                   | To  | From                    | To  |
| 1      | Nil      | Nil           | Nil                        | Nil                       | Nil       | Nil                    | Nil | Nil                     | Nil |

4A.2.1.3: Details of stabilised dumps

| Sl.No. | Dump ID | Number of Terraces | Average Height of Terraces(m) | Lenght of Toe Wall(m) | Lenght of Garland Drain(m) | Area Stablized(Ha) | Method of Stablization |
|--------|---------|--------------------|-------------------------------|-----------------------|----------------------------|--------------------|------------------------|
| 1      | Nil     | Nil                | Nil                           | Nil                   | Nil                        | Nil                | Nil                    |

4A.2.2: Opencast Mining

4A.2.2.1: Bench Parameters

| Pit ID | Year  | Max Height of the Benches in Over Burden (m) | Min Width of the Benches in Over Burden (m) | Slope of the Bench in Over Burden (degree) | Max Height of the Benches in Mineral (m) | Minimum Width of the Benches in Mineral (m) | Slope of the Bench in Mineral (degree) | Overall Slope of Pit (degree) | Number of Benches in Top Soil | Number of Benches in Over Burden | Number of Benches in Mineral | Max Depth of Workings (m) | Depth of Water Table (mRL) | Max Slope Angle of Haul Roads (1xx in) |
|--------|-------|----------------------------------------------|---------------------------------------------|--------------------------------------------|------------------------------------------|---------------------------------------------|----------------------------------------|-------------------------------|-------------------------------|----------------------------------|------------------------------|---------------------------|----------------------------|----------------------------------------|
| Pit-1  | Year1 | 3.50                                         | 6.00                                        | 65.00                                      | 1.50                                     | 6.00                                        | 65.00                                  | 37.00                         | 1                             | 1                                | 1                            | 8.00                      | 80.00                      | 1:16                                   |
| Pit-1  | Year2 | 6.00                                         | 6.00                                        | 65.00                                      | 5.00                                     | 6.00                                        | 65.00                                  | 37.00                         | 1                             | 8                                | 4                            | 50.00                     | 80.00                      | 1:16                                   |
| Pit-1  | Year3 | 6.00                                         | 6.00                                        | 65.00                                      | 5.00                                     | 6.00                                        | 65.00                                  | 37.00                         | 1                             | 8                                | 4                            | 50.00                     | 80.00                      | 1:16                                   |
| Pit-1  | Year4 | 6.00                                         | 6.00                                        | 65.00                                      | 5.00                                     | 6.00                                        | 65.00                                  | 37.00                         | 0                             | 7                                | 3                            | 50.00                     | 80.00                      | 1:16                                   |
| Pit-1  | Year5 | 6.00                                         | 6.00                                        | 65.00                                      | 5.00                                     | 6.00                                        | 65.00                                  | 37.00                         | 0                             | 2                                | 2                            | 50.00                     | 80.00                      | 1:16                                   |

**4A.2.2.2: Yearwise Opencast Development - I Continue**

| Sl.No. | Year  | Pit ID | Bench        | Direction  | Bulk Density of Overburden (BD1) (ton/m <sup>3</sup> ) | Bulk Density of Mineral (BD2) (tonn/m <sup>3</sup> ) | Top Soil Volume (Length x Width x Height) (m <sup>3</sup> ) | Over Burden Volume (Length x Width x Height) (m <sup>3</sup> ) | Over Burden Quantity (t) | ROM Volume (Length x Width x Height) (m <sup>3</sup> ) | ROM Quantity (t) | Recovery | Mineral Reject (t) | Production Main (t) | Production Associated (t) | OB Ratio to Ore (m <sup>3</sup> /ton) |
|--------|-------|--------|--------------|------------|--------------------------------------------------------|------------------------------------------------------|-------------------------------------------------------------|----------------------------------------------------------------|--------------------------|--------------------------------------------------------|------------------|----------|--------------------|---------------------|---------------------------|---------------------------------------|
| 1      | Year1 | Pit-1  | 304 to Above | South West | 2.50                                                   | 2.50                                                 | 8888.90                                                     | 48888.89                                                       | 122222.23                | 13333.34                                               | 33333.35         | 1.00000  | 0.00               | 33333.35            | 0.00                      | 1.4667                                |
| 2      | Year2 | Pit-1  | 304 to Above | North      | 2.50                                                   | 2.50                                                 | 2425.00                                                     | 13337.50                                                       | 33343.75                 | 3637.50                                                | 9093.75          | 1.00000  | 0.00               | 9093.75             | 0.00                      | 1.4667                                |
| 3      | Year2 | Pit-1  | 304 - 298    | North      | 2.50                                                   | 2.50                                                 | 0.00                                                        | 53994.00                                                       | 134985.00                | 0.00                                                   | 0.00             | 1.00000  | 0.00               | 0.00                | 0.00                      | 0.0000                                |
| 4      | Year2 | Pit-1  | 298 - 292    | North      | 2.50                                                   | 2.50                                                 | 0.00                                                        | 41196.00                                                       | 102990.00                | 0.00                                                   | 0.00             | 1.00000  | 0.00               | 0.00                | 0.00                      | 0.0000                                |
| 5      | Year2 | Pit-1  | 292 - 286    | North      | 2.50                                                   | 2.50                                                 | 0.00                                                        | 30120.00                                                       | 75300.00                 | 0.00                                                   | 0.00             | 1.00000  | 0.00               | 0.00                | 0.00                      | 0.0000                                |
| 6      | Year2 | Pit-1  | 286 - 280    | North      | 2.50                                                   | 2.50                                                 | 0.00                                                        | 20778.00                                                       | 51945.00                 | 0.00                                                   | 0.00             | 1.00000  | 0.00               | 0.00                | 0.00                      | 0.0000                                |
| 7      | Year2 | Pit-1  | 280 -274     | North      | 2.50                                                   | 2.50                                                 | 0.00                                                        | 10955.00                                                       | 27387.50                 | 2191.00                                                | 5477.50          | 1.00000  | 0.00               | 5477.50             | 0.00                      | 2.0000                                |
| 8      | Year2 | Pit-1  | 274 - 268    | North      | 2.50                                                   | 2.50                                                 | 0.00                                                        | 1212.00                                                        | 3030.00                  | 6060.00                                                | 15150.00         | 1.00000  | 0.00               | 15150.00            | 0.00                      | 0.0800                                |
| 9      | Year2 | Pit-1  | 268 - 262    | North      | 2.50                                                   | 2.50                                                 | 0.00                                                        | 288.97                                                         | 722.42                   | 1444.94                                                | 3612.33          | 1.00000  | 0.00               | 3612.33             | 0.00                      | 0.0800                                |
| 10     | Year3 | Pit-1  | 304 to above | East       | 2.50                                                   | 2.50                                                 | 2539.00                                                     | 13964.50                                                       | 34911.25                 | 3808.50                                                | 9521.25          | 1.00000  | 0.00               | 9521.25             | 0.00                      | 1.4667                                |
| 11     | Year3 | Pit-1  | 304 - 298    | East       | 2.50                                                   | 2.50                                                 | 0.00                                                        | 13548.00                                                       | 33870.00                 | 0.00                                                   | 0.00             | 1.00000  | 0.00               | 0.00                | 0.00                      | 0.0000                                |
| 12     | Year3 | Pit-1  | 298 - 292    | East       | 2.50                                                   | 2.50                                                 | 0.00                                                        | 11874.00                                                       | 29685.00                 | 0.00                                                   | 0.00             | 1.00000  | 0.00               | 0.00                | 0.00                      | 0.0000                                |
| 13     | Year3 | Pit-1  | 292 - 286    | East       | 2.50                                                   | 2.50                                                 | 0.00                                                        | 10200.00                                                       | 25500.00                 | 0.00                                                   | 0.00             | 1.00000  | 0.00               | 0.00                | 0.00                      | 0.0000                                |
| 14     | Year3 | Pit-1  | 256 - 280    | East       | 2.50                                                   | 2.50                                                 | 0.00                                                        | 8520.00                                                        | 21300.00                 | 0.00                                                   | 0.00             | 1.00000  | 0.00               | 0.00                | 0.00                      | 0.0000                                |
| 15     | Year3 | Pit-1  | 280 -274     | East       | 2.50                                                   | 2.50                                                 | 0.00                                                        | 5700.00                                                        | 14250.00                 | 1140.00                                                | 2850.00          | 1.00000  | 0.00               | 2850.00             | 0.00                      | 2.0000                                |
| 16     | Year3 | Pit-1  | 274 - 268    | East       | 2.50                                                   | 2.50                                                 | 0.00                                                        | 861.00                                                         | 2152.50                  | 4305.00                                                | 10762.50         | 1.00000  | 0.00               | 10762.50            | 0.00                      | 0.0800                                |



|       |       |       |           |            |      |      |      |          |               |         |               |         |      |               |      |        |
|-------|-------|-------|-----------|------------|------|------|------|----------|---------------|---------|---------------|---------|------|---------------|------|--------|
| 17    | Year3 | Pit-1 | 268 - 262 | East       | 2.50 | 2.50 | 0.00 | 815.97   | 2039.92       | 4079.84 | 10199.58      | 1.00000 | 0.00 | 10199.58      | 0.00 | 0.0800 |
| 18    | Year4 | Pit-1 | 304 - 298 | South      | 2.50 | 2.50 | 0.00 | 5202.00  | 13005.00      | 0.00    | 0.00          | 1.00000 | 0.00 | 0.00          | 0.00 | 0.0000 |
| 19    | Year4 | Pit-1 | 298 - 292 | South      | 2.50 | 2.50 | 0.00 | 7020.00  | 17550.00      | 0.00    | 0.00          | 1.00000 | 0.00 | 0.00          | 0.00 | 0.0000 |
| 20    | Year4 | Pit-1 | 292 - 286 | South      | 2.50 | 2.50 | 0.00 | 9396.00  | 23490.00      | 0.00    | 0.00          | 1.00000 | 0.00 | 0.00          | 0.00 | 0.0000 |
| 21    | Year4 | Pit-1 | 286 - 280 | South      | 2.50 | 2.50 | 0.00 | 11016.00 | 27540.00      | 0.00    | 0.00          | 1.00000 | 0.00 | 0.00          | 0.00 | 0.0000 |
| 22    | Year4 | Pit-1 | 280 -274  | South      | 2.50 | 2.50 | 0.00 | 13700.00 | 34250.00      | 2740.00 | 6850.00       | 1.00000 | 0.00 | 6850.00       | 0.00 | 2.0000 |
| 23    | Year4 | Pit-1 | 274 - 268 | South      | 2.50 | 2.50 | 0.00 | 1213.00  | 3032.50       | 6065.00 | 15162.50      | 1.00000 | 0.00 | 15162.50      | 0.00 | 0.0800 |
| 24    | Year4 | Pit-1 | 268 - 262 | South      | 2.50 | 2.50 | 0.00 | 905.67   | 2264.17       | 4528.34 | 11320.83      | 1.00000 | 0.00 | 11320.83      | 0.00 | 0.0800 |
| 25    | Year5 | Pit-1 | 274 - 268 | South West | 2.50 | 2.50 | 0.00 | 1576.00  | 3940.00       | 7880.00 | 19700.00      | 1.00000 | 0.00 | 19700.00      | 0.00 | 0.0800 |
| 26    | Year5 | Pit-1 | 268 - 262 | South West | 2.50 | 2.50 | 0.00 | 1090.67  | 2726.67       | 5453.33 | 13633.33      | 1.00000 | 0.00 | 13633.33      | 0.00 | 0.0800 |
| Total |       |       |           |            |      |      |      |          | 843432.9<br>1 |         | 166666.9<br>2 |         | 0.00 | 166666.9<br>2 | 0.00 |        |

4A.2.2.2 Yearwise Opencast Development - I End

| SI.No. | Year  | Pit ID | Total Topsoil Volume (m³) | Total Over Burden Volume (m³) | Total Over Burden Quantity (t) | Total ROM Volume (m³) | Total ROM Quantity (t) |
|--------|-------|--------|---------------------------|-------------------------------|--------------------------------|-----------------------|------------------------|
| 1      | Year1 | Pit-1  | 8888.90                   | 48888.89                      | 122222.22                      | 13333.33              | 33333.34               |
| 2      | Year2 | Pit-1  | 2425.00                   | 171881.47                     | 429703.67                      | 13333.33              | 33333.34               |
| 3      | Year3 | Pit-1  | 2539.00                   | 65483.47                      | 163708.67                      | 13333.33              | 33333.34               |
| 4      | Year4 | Pit-1  | 0.00                      | 48452.67                      | 121131.67                      | 13333.33              | 33333.34               |
| 5      | Year5 | Pit-1  | 0.00                      | 2666.66                       | 6666.67                        | 13333.33              | 33333.34               |
| Total  |       |        | 13852.90                  | 337373.16                     | 843432.90                      | 66666.65              | 166666.70              |

4A.2.2.3: Transportation & Hauling Equipment

| SI.No. | Type | Make | Capacity (m³) | No. of Equipments |
|--------|------|------|---------------|-------------------|
|--------|------|------|---------------|-------------------|

|   |        |                 |       |   |
|---|--------|-----------------|-------|---|
| 1 | Dumper | TATA(20 Tonnes) | 14.00 | 7 |
|---|--------|-----------------|-------|---|

4A.3: Material Handling Summary

4A.3.1: Studies Undertaken

| Title                              | Study Undertaken | Attachment (only pdf allowed)   |
|------------------------------------|------------------|---------------------------------|
| Blast Vibration Study Report       | Yes              | Nil                             |
| Slope Stability Study Report       | No               | Nil                             |
| Recovery Study Report              | No               | Nil                             |
| Hydrological Study Report          | No               | Nil                             |
| Mineral Beneficiation Study Report | No               | Nil                             |
| Subsidence Study Report            | No               | Nil                             |
| Geotechnical Study Report          | Yes              | <a href="#">Undertaking.pdf</a> |
| Any Other Study Report             | No               | Nil                             |
| Bulk Density Study Report          | Yes              | Nil                             |

4A.3.2: Insitu Mining

| SI.No. | Year  | Waste Quantity(t) | ROM Quantity(t) | Total Handling (t) | ROM Quantity<br>Saleable Mineral (t) | ROM Quantity<br>Mineral Reject (t) | OB Ratio to Ore<br>(Waste Quantity /<br>ROM Quantity ) | Grade Range (%)    |
|--------|-------|-------------------|-----------------|--------------------|--------------------------------------|------------------------------------|--------------------------------------------------------|--------------------|
| 1      | Year1 | 122222.22         | 33333.33        | 155555.55          | 33333.33                             | 0.00                               | 3.67                                                   | Ca-49.73, MgO-1.86 |
| 2      | Year2 | 429703.67         | 33333.33        | 463037.00          | 33333.33                             | 0.00                               | 12.89                                                  | Ca-49.73, MgO-1.86 |
| 3      | Year3 | 163708.67         | 33333.33        | 197042.00          | 33333.33                             | 0.00                               | 4.91                                                   | Ca-49.73, MgO-1.86 |
| 4      | Year4 | 121131.67         | 33333.33        | 154465.00          | 33333.33                             | 0.00                               | 3.63                                                   | Ca-49.73, MgO-     |

|   |       |           |           |            |           |      |      |                    |
|---|-------|-----------|-----------|------------|-----------|------|------|--------------------|
|   |       |           |           |            |           |      |      | 1.86               |
| 5 | Year5 | 6666.66   | 33333.33  | 39999.99   | 33333.33  | 0.00 | 0.20 | Ca-49.73, MgO-1.86 |
|   | Total | 843432.89 | 166666.65 | 1010099.54 | 166666.65 | 0.00 |      |                    |

4A.3.3: Dump workings

| Sl.No. | Year | Dump ID | Latitude (dd:mm:ss.ss) |     | Longitude (dd:mm:ss.ss) |     | Area (m2) | Avg Height of Dump (m) | Volume (m³) | Total Dump Quantity (t) | Proposed Dump Handling Quantity (t) (A) | Proposed Recovery of Saleable Mineral (t)(B) | Proposed Waste Quantity (t) (A-B) | Grade Range (%) | Justification |
|--------|------|---------|------------------------|-----|-------------------------|-----|-----------|------------------------|-------------|-------------------------|-----------------------------------------|----------------------------------------------|-----------------------------------|-----------------|---------------|
|        |      |         | From                   | To  | From                    | To  |           |                        |             |                         |                                         |                                              |                                   |                 |               |
| 1      | Nil  | Nil     | Nil                    | Nil | Nil                     | Nil | Nil       | Nil                    | Nil         | Nil                     | Nil                                     | Nil                                          | Nil                               | Nil             | Nil           |

4A.3.4: Calculation Summary

| Year                                              | Year1    | Year2    | Year3    | Year4    | Year5    | Total     |
|---------------------------------------------------|----------|----------|----------|----------|----------|-----------|
| (A) Total ROM quantity (t)                        | 33333.33 | 33333.33 | 33333.33 | 33333.33 | 33333.33 | 166666.65 |
| (B) Saleable ore from ROM (t)                     | 33333.33 | 33333.33 | 33333.33 | 33333.33 | 33333.33 | 166666.65 |
| (C) Proposed Dump Handling Quantity (t)           | 0.00     | 0.00     | 0.00     | 0.00     | 0.00     | 0.00      |
| (D) Saleable Ore recovered from dump workings (t) | 0.00     | 0.00     | 0.00     | 0.00     | 0.00     | 0.00      |
| (E) Total Saleable Ore (t)(=B+D)                  | 33333.33 | 33333.33 | 33333.33 | 33333.33 | 33333.33 | 166666.65 |
| (F) Total Quantity Handled (t)(=A+C)              | 33333.33 | 33333.33 | 33333.33 | 33333.33 | 33333.33 | 166666.65 |

4A.4: Machine Calculation

4A.4.1: Machine Requirement Summary

|                                                                     |               |
|---------------------------------------------------------------------|---------------|
| Number of Average Working Days in One Year (A)                      | 300           |
| Number of Shifts per Day (B)                                        | 1             |
| Material Handling Required per Day (t) ((D)=Largest of (Q1,Q5)/(A)) | 1543.45       |
| Material to be Handled per Shift (t) ((E)=(D)/(B))                  | 1543.45       |
| Handling Required per Hour (t) ((F)=(E)/8 hours)                    | 257.17        |
| Effective Shift Time                                                | 6 hrs 00 mins |

4A.4.2: Shovel / Excavator Requirement

| Effective Shift Time |                     |                         |                        |                  | 6 hrs                     |                                    |                    |                      |                                                         | 00 mins                                                                                |                                                  |                                                                                  |                                                     |                       |
|----------------------|---------------------|-------------------------|------------------------|------------------|---------------------------|------------------------------------|--------------------|----------------------|---------------------------------------------------------|----------------------------------------------------------------------------------------|--------------------------------------------------|----------------------------------------------------------------------------------|-----------------------------------------------------|-----------------------|
| SI.No.               | Type                | Bucket Capacity (m³)(A) | Bucket Fill Factor (B) | Swell Factor (C) | Tonnage Factor (t/m³) (D) | Machine Utilization Factor (%) (U) | Efficiency (%) (E) | Cycle time (sec) (F) | (G) TPH =TPH (G) =((3600 x A x B x C x D x E x U ) / F) | Total Hours (H) =Number of working days x Number of shifts/day x Effective shift hours | Yearly handling by one Excavator (t) (I)=(G x H) | Maximum handling of the material by this machine during the block period (t) (J) | Number of excavator machines required (K) = (J / I) | Standby excavator (L) |
| 1                    | Hydraulic excavator | 1.80                    | 0.8                    | 0.7              | 2.50                      | 0.80                               | 0.80               | 50                   | 116.12                                                  | 1800                                                                                   | 209016.00                                        | 463037.50                                                                        | 2.22                                                | 1                     |

4A.4.3: Dumper Requirement

| Effective Shift Time |                                                                                                                        |                             |                                |                         | 6 hrs                                                      |                                            |                                                    |                                                      |                               | 00 mins                                   |                                              |                                                                                  |                                           |                           |
|----------------------|------------------------------------------------------------------------------------------------------------------------|-----------------------------|--------------------------------|-------------------------|------------------------------------------------------------|--------------------------------------------|----------------------------------------------------|------------------------------------------------------|-------------------------------|-------------------------------------------|----------------------------------------------|----------------------------------------------------------------------------------|-------------------------------------------|---------------------------|
| SI.No.               | Total Hours=Number of working days (W)x Number of shifts/day x Effective shift hours (Machine Requirement Summary) (A) | Capacity of Dumpers (t) (B) | Speed of the dumper (KMPH) (i) | Lead Distance (KM) (ii) | Time taken to cover distance in minutes(iii) = (ii/i) x 60 | Queuing, Loading Time at Shovel (min) (iv) | Queuing, Unloading Time during unloading (min) (v) | Total Time to complete one trip(vi) = (iii + iv + v) | No. of Trips / hr = (60 / vi) | Total transportation per hour =( B X vii) | Yearly handling by one dumper (ix) = A x TPH | Maximum handling of the material by this machine during the block period (t) (x) | Number of dumpers will be (xi) =( x / ix) | Plus Standby dumper (xii) |
| 1                    | 1800                                                                                                                   | 14.00                       | 10.00                          | 2.00                    | 12.00                                                      | 6.00                                       | 3.00                                               | 21.00                                                | 2                             | 40                                        | 72072.00                                     | 463036.00                                                                        | 6                                         | 1                         |

#### 4A.4.4: Drill Machine Requirement

| Effective Shift Time |               |                                                |             |            | 6 hrs                                     |                                             |                    |                                                                                            |                         | 00 mins                                                                                                                          |                                   |                                                                                             |                                                                                                                                              |                |
|----------------------|---------------|------------------------------------------------|-------------|------------|-------------------------------------------|---------------------------------------------|--------------------|--------------------------------------------------------------------------------------------|-------------------------|----------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|---------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| Sl.No.               | Type of Drill | Depth of Hole(including Sub-grade Drilling (m) | Spacing (m) | Burden (m) | Bulk Density of Waste (t/m <sup>3</sup> ) | Bulk Density of Mineral (t/m <sup>3</sup> ) | Yield per Hole (t) | Yield per Meter (t/m) = Yield per Hole (t)/Depth of Hole(including Sub-grade Drilling (m)) | Annual Target Known (t) | Drilling Requirement per Day (m) = (Annual Target Known (t) / Yield per Meter (t/m) )/Number of Average Working Days in One Year | Drilling Requirement per Shift(m) | Rate of Drilling per Hours (m/hr) = Drilling Requirement per Shift (m)/Effective Shift Time | Required No. of drills (m/c) = Required rate of drilling in meters per hr./ Actual rate of drilling in meters per hr of the machine deployed | Stand by Drill |

|   |                                       |      |      |      |      |      |        |       |           |       |       |      |      |   |
|---|---------------------------------------|------|------|------|------|------|--------|-------|-----------|-------|-------|------|------|---|
|   |                                       |      |      |      |      |      |        |       |           | (A)   |       |      |      |   |
| 1 | In-the-hole<br>longhole<br>drill rigs | 6.60 | 4.00 | 3.00 | 2.50 | 2.50 | 198.00 | 30.00 | 463037.50 | 51.44 | 51.44 | 8.57 | 1.00 | 1 |

#### 4A.4.5: Machine Deployment Details

#### 4A.4.5.1: Excavator & Loading Equipment

| Sl.No. | Type               | Make | Capacity (m <sup>3</sup> ) | No. of Equipments |
|--------|--------------------|------|----------------------------|-------------------|
| 1      | HydraulicExcavator | L&T  | 1.80                       | 3                 |

#### 4A.4.5.2: Dozers Details

| SI.No. | Type     | Make     | Capacity (hp) | No. of Equipments |
|--------|----------|----------|---------------|-------------------|
| 1      | Straight | BEML-155 | 320.00        | 1                 |

#### 4A.4.5.3: Drilling Details

| SI.No. | Type | Make       | Capacity (m/hr) | Diameter of Hole(mm) |
|--------|------|------------|-----------------|----------------------|
| 1      | DTH  | ATLASCOPCO | 15.00           | 110.00               |

#### 4A.5 Blasting Requirement

#### 4A.5.1: Blasting & Explosive Requirement in Waste/Development

| Sl.No. | Drill Pattern /<br>Spacing of<br>Holes (m) | Burden of<br>Holes (m) | Number of<br>Rows / Rings | Yield per<br>Holes in<br>Waste (m <sup>3</sup> ) | Frequency of<br>Blasting in a<br>Week | Maximum<br>Number of<br>Holes Blasted<br>in a Round | Charge per<br>Hole (kg) | Charge per<br>Round (kg) | Explosive<br>Requirement<br>Per Month in<br>Development<br>(kg) | Powder Factor<br>in<br>Development /<br>Waste (t/kg) | Depth Of<br>Hole |
|--------|--------------------------------------------|------------------------|---------------------------|--------------------------------------------------|---------------------------------------|-----------------------------------------------------|-------------------------|--------------------------|-----------------------------------------------------------------|------------------------------------------------------|------------------|
|--------|--------------------------------------------|------------------------|---------------------------|--------------------------------------------------|---------------------------------------|-----------------------------------------------------|-------------------------|--------------------------|-----------------------------------------------------------------|------------------------------------------------------|------------------|



|   |   |   |   |    |   |   |    |     |      |      |   |
|---|---|---|---|----|---|---|----|-----|------|------|---|
| 1 | 4 | 3 | 2 | 72 | 6 | 8 | 25 | 200 | 5000 | 7.16 | 6 |
|---|---|---|---|----|---|---|----|-----|------|------|---|

4A.5.2: Blasting & Explosive Requirement in Mineral / Ore

|                                   |                                                                    |                                                                 |                               |                              |                       |                                                    |                                              |                                                                   |                                                                 |                                                      |                                |                                                                                          |                                      |                                             |                                                 |                           |              |                                               |                     |
|-----------------------------------|--------------------------------------------------------------------|-----------------------------------------------------------------|-------------------------------|------------------------------|-----------------------|----------------------------------------------------|----------------------------------------------|-------------------------------------------------------------------|-----------------------------------------------------------------|------------------------------------------------------|--------------------------------|------------------------------------------------------------------------------------------|--------------------------------------|---------------------------------------------|-------------------------------------------------|---------------------------|--------------|-----------------------------------------------|---------------------|
| Type of Explosive                 |                                                                    |                                                                 |                               |                              |                       |                                                    |                                              |                                                                   |                                                                 | Type of Explosives used / to be Used                 |                                |                                                                                          |                                      |                                             |                                                 |                           |              |                                               |                     |
| Ammonium Nitrate Fuel Oil Mixture |                                                                    |                                                                 |                               |                              |                       |                                                    |                                              |                                                                   |                                                                 | Aluminised Gelled Slurry Explosives (Large Diameter) |                                |                                                                                          |                                      |                                             |                                                 |                           |              |                                               |                     |
| SI.No.                            | Total ROM p<br>roposed<br>to be<br>handled<br>in CU<br>M/annu<br>m | Total ROM p<br>roposed<br>to be<br>handled<br>in<br>CUM/<br>day | Spacing<br>of<br>Holes<br>(m) | Burden<br>of<br>Holes<br>(m) | Numbe<br>r of<br>Rows | Yield<br>per<br>Holes<br>in<br>ROM<br>Zone<br>(m³) | Freque<br>ncy of<br>Blasting<br>in a<br>Week | Maxim<br>um<br>Numbe<br>r of<br>Holes<br>Blasted<br>in a<br>Round | No of<br>Holes R<br>equired<br>to be<br>Blasted<br>per<br>Round | Charge<br>per<br>Hole<br>(kg)                        | Charge<br>per<br>Round<br>(kg) | Explosi<br>ve Req<br>uireme<br>nt Per<br>Month<br>for<br>ROM<br>Zone<br>Blasting<br>(kg) | Powder<br>Factor<br>in Ore<br>(t/kg) | Pop<br>Shootin<br>g (no of<br>Boulder<br>s) | Plaster<br>Shootin<br>g (no of<br>Boulder<br>s) | Use of<br>Rockbr<br>eaker | Capacit<br>y | Second<br>ary<br>Blasting<br>Require<br>ments | Depth<br>Of<br>Hole |
| 1                                 | 13334                                                              | 44.44                                                           | 4                             | 3                            | 5                     | 72                                                 | 1                                            | 4                                                                 | 4                                                               | 25                                                   | 100                            | 400                                                                                      | 6.9                                  | 0                                           | 0                                               | Yes                       | 133          | 0                                             | 6                   |

4A.6: Man Power Deployment

4A.6.1: Managerial

|        |                 |                                 |                                 |                                 |                                       |                                 |
|--------|-----------------|---------------------------------|---------------------------------|---------------------------------|---------------------------------------|---------------------------------|
| SI.No. | Particular      | Number of Persons in<br>Shift 1 | Number of Persons in<br>Shift 2 | Number of Persons in<br>Shift 3 | Number of Persons in<br>General Shift | Total No. of Persons per<br>day |
| 1      | 1st Class       | 0                               | 0                               | 0                               | 1                                     | 1                               |
| 2      | Geologist       | 0                               | 0                               | 0                               | 1                                     | 1                               |
| 3      | Mining Engineer | 0                               | 0                               | 0                               | 1                                     | 1                               |

4A.6.2: Supervisory

|        |            |                      |                      |                      |                      |                          |
|--------|------------|----------------------|----------------------|----------------------|----------------------|--------------------------|
| SI.No. | Particular | Number of Persons in | Number of Persons in | Number of Persons in | Number of Persons in | Total No. of Persons per |
|--------|------------|----------------------|----------------------|----------------------|----------------------|--------------------------|

|   |           |         |         |         |               |     |
|---|-----------|---------|---------|---------|---------------|-----|
|   |           | Shift 1 | Shift 2 | Shift 3 | General Shift | day |
| 1 | Foreman   | 0       | 0       | 0       | 1             | 1   |
| 2 | Mine-mate | 0       | 0       | 0       | 1             | 1   |

4A.6.3: Skilled Workers / Operators

| SI.No. | Particular            | Number of Persons in Shift 1 | Number of Persons in Shift 2 | Number of Persons in Shift 3 | Number of Persons in General Shift | Total No. of Persons per day |
|--------|-----------------------|------------------------------|------------------------------|------------------------------|------------------------------------|------------------------------|
| 1      | Dozer/Grader Operator | 0                            | 0                            | 0                            | 1                                  | 1                            |
| 2      | Dumper Operator       | 0                            | 0                            | 0                            | 4                                  | 4                            |
| 3      | Drill Operator        | 0                            | 0                            | 0                            | 2                                  | 2                            |
| 4      | Operator              | 0                            | 0                            | 0                            | 2                                  | 2                            |
| 5      | Other                 | 0                            | 0                            | 0                            | 1                                  | 1                            |

4A.6.4: Semi-skilled Workers

| SI.No. | Number of Persons in Shift 1 | Number of Persons in Shift 2 | Number of Persons in Shift 3 | Number of Persons in General Shift | Total No. of Persons per day |
|--------|------------------------------|------------------------------|------------------------------|------------------------------------|------------------------------|
| 1      | 0                            | 0                            | 0                            | 5                                  | 5                            |

4A.6.5: Unskilled Workers

| SI.No. | Number of Persons in Shift 1 | Number of Persons in Shift 2 | Number of Persons in Shift 3 | Number of Persons in General Shift | Total No. of Persons per day |
|--------|------------------------------|------------------------------|------------------------------|------------------------------------|------------------------------|
| 1      | 0                            | 0                            | 0                            | 2                                  | 2                            |

4A.6.6: Others Specify

| SI.No. | Particular | Number of Persons in | Number of Persons in | Number of Persons in | Number of Persons in | Total No. of Persons per |
|--------|------------|----------------------|----------------------|----------------------|----------------------|--------------------------|
|--------|------------|----------------------|----------------------|----------------------|----------------------|--------------------------|



|   |                     |         |         |         |               |     |
|---|---------------------|---------|---------|---------|---------------|-----|
|   |                     | Shift 1 | Shift 2 | Shift 3 | General Shift | day |
| 1 | Mechanical Engineer | 0       | 0       | 0       | 1             | 1   |

4A.6.7: No of Persons Engaged Per Day

| Sl.No.                                                                            | Number of Persons in Shift 1 | Number of Persons in Shift 2 | Number of Persons in Shift 3 | Number of Persons in General Shift | Total No. of Persons per day |
|-----------------------------------------------------------------------------------|------------------------------|------------------------------|------------------------------|------------------------------------|------------------------------|
| 1                                                                                 | 22                           | 0                            | 0                            | 0                                  | 22                           |
| No of Shifts per Day ((A) = Machine Requirement Summary (B))                      |                              |                              | 1                            |                                    |                              |
| Average Daily Employment per Shift ((B) = (Total Number of Person per Day) / (A)) |                              |                              | 22                           |                                    |                              |
| Material to be Handled per Shift ((C) = Machine Requirement Summary (E))          |                              |                              | 1543                         |                                    |                              |

4A.6.8: Supervision

| Sl.No. | Particular    | Qualification         | Requirement / Proposed | In Position / Existing Strength | (Requirement / Proposed) - (In Position / Existing Strength) = (-) Shortage / (+) Excess | Remarks                                               |
|--------|---------------|-----------------------|------------------------|---------------------------------|------------------------------------------------------------------------------------------|-------------------------------------------------------|
| 1      | Mines Manager | 1st Class Certificate | 1                      | 0                               | 1                                                                                        | ManPowerWillBe DeployedWhenMining OperationCommences. |
| 2      | Mine Foreman  | Diploma(Mining)       | 1                      | 0                               | 1                                                                                        | ManPowerWillBe DeployedWhenMining OperationCommences. |
| 3      | Geologist     | M.Sc. Geology         | 1                      | 0                               | 1                                                                                        | ManPowerWillBe DeployedWhenMining OperationCommences. |

4A.7: Waste Management

4A.7.1: Existing Dump

| Sl.No. | Year | Dump Id | Type of Dump | Proposed Area (ha) | Height (m) | Latitude (dd:mm:ss.ss) |     | Longitude (dd:mm:ss.ss) |     | Total Dump Quantity (m <sup>3</sup> ) | Existing Dump Location |
|--------|------|---------|--------------|--------------------|------------|------------------------|-----|-------------------------|-----|---------------------------------------|------------------------|
|        |      |         |              |                    |            | From                   | To  | From                    | To  |                                       |                        |
| 1      | Nil  | Nil     | Nil          | Nil                | Nil        | Nil                    | Nil | Nil                     | Nil | Nil                                   | Nil                    |

4A.7.2: New Dump

| Sl.No. | Year  | Dump Id | Type of Dump | Proposed Area (ha) | Height (m) | Latitude (dd:mm:ss.ss) |             | Longitude (dd:mm:ss.ss) |             | Total Dump Quantity (m <sup>3</sup> ) | New Dump Location |
|--------|-------|---------|--------------|--------------------|------------|------------------------|-------------|-------------------------|-------------|---------------------------------------|-------------------|
|        |       |         |              |                    |            | From                   | To          | From                    | To          |                                       |                   |
| 1      | Year1 | D1      | Waste        | 0.50               | 10.00      | 26:58:09.23            | 26:58:12.01 | 73:29:20.78             | 73:29:16.90 | 50220.00                              | Near T1 pillar    |
| 2      | Year2 | D1      | Waste        | 1.15               | 15.00      | 26:58:12.01            | 26:58:12.79 | 73:29:16.90             | 73:29:16.90 | 173214.00                             | Near T1pillar     |
| 3      | Year3 | D1      | Waste        | 1.01               | 7.00       | 26:58:15.27            | 26:58:12.96 | 73:29:21.14             | 73:29:21.22 | 71043.00                              | Near T2pillar     |
| 4      | Year4 | D1      | Waste        | 0.67               | 7.00       | 26:58:15.31            | 26:58:12.96 | 73:29:24.37             | 73:29:24.41 | 47110.00                              | Near T2pillar     |

4A.7.3: Existing Stack

| Sl.No. | Year | Stack ID | Type of Stack | Proposed Area (ha) | Height (m) | Latitude (dd:mm:ss.ss) |     | Longitude (dd:mm:ss.ss) |     | Total Stack Quantity (m <sup>3</sup> ) | Existing Stack Location |
|--------|------|----------|---------------|--------------------|------------|------------------------|-----|-------------------------|-----|----------------------------------------|-------------------------|
|        |      |          |               |                    |            | From                   | To  | From                    | To  |                                        |                         |
| 1      | Nil  | Nil      | Nil           | Nil                | Nil        | Nil                    | Nil | Nil                     | Nil | Nil                                    | Nil                     |

4A.7.4: New Stack

| Sl.No. | Year  | Stack ID | Type of Stack     | Proposed Area (ha) | Height (m) | Latitude (dd:mm:ss.ss) |             | Longitude (dd:mm:ss.ss) |             | Total Stack Quantity (m <sup>3</sup> ) | New Stack Location |
|--------|-------|----------|-------------------|--------------------|------------|------------------------|-------------|-------------------------|-------------|----------------------------------------|--------------------|
|        |       |          |                   |                    |            | From                   | To          | From                    | To          |                                        |                    |
| 1      | Year1 | S1       | Stack for mineral | 0.02               | 4.00       | 26:58:12.11            | 26:58:11.91 | 73:29:21.64             | 73:29:22.28 | 648.00                                 | Near Working Pit   |
| 2      | Year2 | S1       | Stack for mineral | 0.02               | 4.00       | 26:58:12.11            | 26:58:11.91 | 73:29:21.64             | 73:29:22.28 | 648.00                                 | Near Working Pit   |
| 3      | Year3 | S1       | Stack for mineral | 0.02               | 4.00       | 26:58:12.11            | 26:58:11.91 | 73:29:21.64             | 73:29:22.28 | 648.00                                 | Near Working Pit   |

|   |       |    |                   |      |      |             |             |             |             |        |                  |
|---|-------|----|-------------------|------|------|-------------|-------------|-------------|-------------|--------|------------------|
| 4 | Year4 | S1 | Stack for mineral | 0.02 | 4.00 | 26:58:12.11 | 26:58:11.91 | 73:29:21.64 | 73:29:22.28 | 648.00 | Near Working Pit |
| 5 | Year5 | S1 | Stack for mineral | 0.02 | 4.00 | 26:58:12.11 | 26:58:11.91 | 73:29:21.64 | 73:29:22.28 | 648.00 | Near Working Pit |

4A.8: Mineral Waste Handling To Utilize As Minor Mineral

| Sl.No. | Year  | Dump ID | Type of Dump | Proposed Area (ha) | Quantity Handled (t) | Quantity Recovered (t) | Name Of Minor Mineral | Alternative Waste Utilization (m³) |
|--------|-------|---------|--------------|--------------------|----------------------|------------------------|-----------------------|------------------------------------|
| 1      | Year1 | D1      | Waste        | 0.50               | 125550.00            | 62775.00               | Masonry Stone         | 25110.00                           |
| 2      | Year2 | D1      | Waste        | 1.49               | 432975.00            | 216487.50              | Masonry Stone         | 89595.00                           |
| 3      | Year3 | D1      | Waste        | 1.01               | 177607.50            | 88803.75               | Masonry Stone         | 35521.50                           |
| 4      | Year4 | D1      | Waste        | 0.47               | 93037.50             | 41518.75               | Masonry Stone         | 16607.50                           |

4A.9: Use of Minerals

| Sl.No. | Proposed Use Of Mineral | Name Of Mineral | Relevant Use Of Mineral                                                       | Physical Specifications                                                                                                          | Chemical Specifications |
|--------|-------------------------|-----------------|-------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|-------------------------|
| 1      | Direct Selling          | LIMESTONE       | Manufacturing of quick lime and hydrated lime and also use in Cement industry | Lumps size upto1200 mm, Colour of limestone is white to gray, Hardness of limestone is 3 on Mohs scale. The bulk density is 2.5. | CaO49.73%, MgO1.86%     |

\* Choose among these:

1. Captive use in own industry

2. Direct Selling

3. Selling Post-Beneficiation /Up-gradation

\*Select more than one, if applicable

# **Chapter 4 B : Mining Operations UG : NA**

Approved

Chapter 5: Sustainable Mining

5.1: Sustainable Mining and SDF Implementations in Compliance of Rule 35 of MCDR’2017

The block holder shall monitor his mining and allied activities as per the template of Star Rating in the format specified in this behalf by the Indian Bureau of mines from time to time, and shall submit online itsself –assessment report before the 1stdayof July every year for the previous financial year, along with the digital images of mining lease block under rule34A, to the Regional controller or the authorized officer of the Indian Bureau of Mines.

(Total 200 characters)

|                                                                                             |             |
|---------------------------------------------------------------------------------------------|-------------|
| Compliance of Vishakha Committee Guidelines for prevention of women harassment at workplace | Implemented |
|---------------------------------------------------------------------------------------------|-------------|

5.2: CSR INITIATIVES

5.2.1: Year1

| Details of Work Proposed during the Year / Measures Planned for the Affected Segment | Cumulative Work done / Measures Taken |
|--------------------------------------------------------------------------------------|---------------------------------------|
| 5.2.1.1: Area to be Developed for Recreation                                         |                                       |
| Area (Ha)                                                                            | Area (Ha)                             |
| Nil                                                                                  | Nil                                   |
| 5.2.1.2: Area for Water Storage & Recharge Facility                                  |                                       |
| Area (Ha)                                                                            | Area (Ha)                             |
| Nil                                                                                  | Nil                                   |
| 5.2.1.3: Efforts Made towards Housing for Local Communities                          |                                       |
| Number of Houses                                                                     | Number of Houses                      |
| Nil                                                                                  | Nil                                   |

| 5.2.1.4: Efforts Made towards Providing Transport to Local Communities                         |                                                 |                         |
|------------------------------------------------------------------------------------------------|-------------------------------------------------|-------------------------|
| Number of Beneficiaries                                                                        |                                                 | Number of Beneficiaries |
| Nil                                                                                            |                                                 | Nil                     |
| 5.2.1.5: Efforts Made towards Providing Healthcare to Local Communities                        |                                                 |                         |
| Number of Beneficiaries                                                                        |                                                 | Number of Beneficiaries |
| 5                                                                                              |                                                 | 5                       |
| 5.2.1.6: Efforts Made towards Providing Hygiene & Sanitation to Local Communities              |                                                 |                         |
| Number of Beneficiaries                                                                        |                                                 | Number of Beneficiaries |
| 5                                                                                              |                                                 | 5                       |
| 5.2.1.7: Efforts Made towards Skill Development Programs to Local Communities                  |                                                 |                         |
| Number of Beneficiaries                                                                        |                                                 | Number of Beneficiaries |
| 5                                                                                              |                                                 | 5                       |
| 5.2.1.8: Efforts Made to Promote Education & Knowledge Based Initiatives                       |                                                 |                         |
| Number of Beneficiaries                                                                        |                                                 | Number of Beneficiaries |
| 5                                                                                              |                                                 | 5                       |
| 5.2.1.9: Communication Facilities Provided to Local Communities                                |                                                 |                         |
| Number of Beneficiaries                                                                        |                                                 | Number of Beneficiaries |
| 0                                                                                              |                                                 | 0                       |
| 5.2.1.10: Any Other Steps Taken for Improving the Socio-Economic Standard of Local Communities |                                                 |                         |
| Number of Beneficiaries                                                                        |                                                 | Number of Beneficiaries |
| 5                                                                                              |                                                 | 5                       |
| 5.2.1.11: Adoption of ODF                                                                      |                                                 |                         |
| Number of Toilets Built inside the Lease Area                                                  | Number of Toilets Built outside the Lease Area: | Number of Beneficiaries |
| 1                                                                                              | 0                                               | 0                       |



| 5.2.1.12: Awareness Program among Mine Workers for Swatchata |                                       |
|--------------------------------------------------------------|---------------------------------------|
| Number of Swatchata Programmes Proposed                      | Number of Swatchata Programmes Held   |
| 1                                                            | 1                                     |
| 5.2.1.13: Efforts for green energy                           |                                       |
| Total energy consumption (KWh)                               | Green energy consumption (% of total) |
| 50.00                                                        | 10.00                                 |
| 5.2.1.14: Water & recycled use                               |                                       |
| Total water consumption (KLD)                                | Water recycled (% of total)           |
| 31.70                                                        | 0.00                                  |

5.2.2: Year2

| Details of Work Proposed during the Year / Measures Planned for the Affected Segment | Cumulative Work done / Measures Taken |
|--------------------------------------------------------------------------------------|---------------------------------------|
| 5.2.2.1: Area to be Developed for Recreation                                         |                                       |
| Area (Ha)                                                                            | Area (Ha)                             |
| 0.00                                                                                 | 0.00                                  |
| 5.2.2.2: Area for Water Storage & Recharge Facility                                  |                                       |
| Area (Ha)                                                                            | Area (Ha)                             |
| 0.00                                                                                 | 0.00                                  |
| 5.2.2.3: Efforts Made towards Housing for Local Communities                          |                                       |
| Number of Houses                                                                     | Number of Houses                      |
| 0                                                                                    | 0                                     |
| 5.2.2.4: Efforts Made towards Providing Transport to Local Communities               |                                       |
| Number of Beneficiaries                                                              | Number of Beneficiaries               |
| 0                                                                                    | 0                                     |

**5.2.2.5: Efforts Made towards Providing Healthcare to Local Communities**

| Number of Beneficiaries | Number of Beneficiaries |
|-------------------------|-------------------------|
| 10                      | 15                      |

**5.2.2.6: Efforts Made towards Providing Hygiene & Sanitation to Local Communities**

| Number of Beneficiaries | Number of Beneficiaries |
|-------------------------|-------------------------|
| 5                       | 10                      |

**5.2.2.7: Efforts Made towards Skill Development Programs to Local Communities**

| Number of Beneficiaries | Number of Beneficiaries |
|-------------------------|-------------------------|
| 5                       | 10                      |

**5.2.2.8: Efforts Made to Promote Education & Knowledge Based Initiatives**

| Number of Beneficiaries | Number of Beneficiaries |
|-------------------------|-------------------------|
| 10                      | 15                      |

**5.2.2.9: Communication Facilities Provided to Local Communities**

| Number of Beneficiaries | Number of Beneficiaries |
|-------------------------|-------------------------|
| 0                       | 0                       |

**5.2.2.10: Any Other Steps Taken for Improving the Socio-Economic Standard of Local Communities**

| Number of Beneficiaries | Number of Beneficiaries |
|-------------------------|-------------------------|
| 10                      | 15                      |

**5.2.2.11: Adoption of ODF**

| Number of Toilets Built inside the Lease Area | Number of Toilets Built outside the Lease Area: | Number of Beneficiaries |
|-----------------------------------------------|-------------------------------------------------|-------------------------|
| 0                                             | 0                                               | 0                       |

**5.2.2.12: Awareness Program among Mine Workers for Swatchata**

| Number of Swatchata Programmes Proposed | Number of Swatchata Programmes Held |
|-----------------------------------------|-------------------------------------|
| 1                                       | 2                                   |



| 5.2.2.13: Efforts for green energy |                                       |
|------------------------------------|---------------------------------------|
| Total energy consumption (KWh)     | Green energy consumption (% of total) |
| 50.00                              | 20.00                                 |

| 5.2.2.14: Water & recycled use |                             |
|--------------------------------|-----------------------------|
| Total water consumption (KLD)  | Water recycled (% of total) |
| 41.70                          | 0.00                        |

5.2.3: Year3

| Details of Work Proposed during the Year / Measures Planned for the Affected Segment | Cumulative Work done / Measures Taken |
|--------------------------------------------------------------------------------------|---------------------------------------|
| 5.2.3.1: Area to be Developed for Recreation                                         |                                       |
| Area (Ha)                                                                            | Area (Ha)                             |
| 0.00                                                                                 | 0.00                                  |

| 5.2.3.2: Area for Water Storage & Recharge Facility |           |
|-----------------------------------------------------|-----------|
| Area (Ha)                                           | Area (Ha) |
| 0.00                                                | 0.00      |

| 5.2.3.3: Efforts Made towards Housing for Local Communities |                  |
|-------------------------------------------------------------|------------------|
| Number of Houses                                            | Number of Houses |
| 0                                                           | 0                |

| 5.2.3.4: Efforts Made towards Providing Transport to Local Communities |                         |
|------------------------------------------------------------------------|-------------------------|
| Number of Beneficiaries                                                | Number of Beneficiaries |
| 0                                                                      | 0                       |

| 5.2.3.5: Efforts Made towards Providing Healthcare to Local Communities |                         |
|-------------------------------------------------------------------------|-------------------------|
| Number of Beneficiaries                                                 | Number of Beneficiaries |
| 10                                                                      | 25                      |

| 5.2.3.6: Efforts Made towards Providing Hygiene & Sanitation to Local Communities              |                                                 |                                       |
|------------------------------------------------------------------------------------------------|-------------------------------------------------|---------------------------------------|
| Number of Beneficiaries                                                                        |                                                 | Number of Beneficiaries               |
| 5                                                                                              |                                                 | 15                                    |
| 5.2.3.7: Efforts Made towards Skill Development Programs to Local Communities                  |                                                 |                                       |
| Number of Beneficiaries                                                                        |                                                 | Number of Beneficiaries               |
| 10                                                                                             |                                                 | 25                                    |
| 5.2.3.8: Efforts Made to Promote Education & Knowledge Based Initiatives                       |                                                 |                                       |
| Number of Beneficiaries                                                                        |                                                 | Number of Beneficiaries               |
| 10                                                                                             |                                                 | 25                                    |
| 5.2.3.9: Communication Facilities Provided to Local Communities                                |                                                 |                                       |
| Number of Beneficiaries                                                                        |                                                 | Number of Beneficiaries               |
| 0                                                                                              |                                                 | 0                                     |
| 5.2.3.10: Any Other Steps Taken for Improving the Socio-Economic Standard of Local Communities |                                                 |                                       |
| Number of Beneficiaries                                                                        |                                                 | Number of Beneficiaries               |
| 15                                                                                             |                                                 | 30                                    |
| 5.2.3.11: Adoption of ODF                                                                      |                                                 |                                       |
| Number of Toilets Built inside the Lease Area                                                  | Number of Toilets Built outside the Lease Area: | Number of Beneficiaries               |
| 0                                                                                              | 0                                               | 0                                     |
| 5.2.3.12: Awareness Program among Mine Workers for Swatchata                                   |                                                 |                                       |
| Number of Swatchata Programmes Proposed                                                        |                                                 | Number of Swatchata Programmes Held   |
| 1                                                                                              |                                                 | 3                                     |
| 5.2.3.13: Efforts for green energy                                                             |                                                 |                                       |
| Total energy consumption (KWh)                                                                 |                                                 | Green energy consumption (% of total) |
| 50.00                                                                                          |                                                 | 30.00                                 |

| 5.2.3.14: Water & recycled use |                             |
|--------------------------------|-----------------------------|
| Total water consumption (KLD)  | Water recycled (% of total) |
| 51.70                          | 0.00                        |

5.2.4: Year4

| Details of Work Proposed during the Year / Measures Planned for the Affected Segment | Cumulative Work done / Measures Taken |
|--------------------------------------------------------------------------------------|---------------------------------------|
| 5.2.4.1: Area to be Developed for Recreation                                         |                                       |
| Area (Ha)                                                                            | Area (Ha)                             |
| 0.00                                                                                 | 0.00                                  |
| 5.2.4.2: Area for Water Storage & Recharge Facility                                  |                                       |
| Area (Ha)                                                                            | Area (Ha)                             |
| 0.00                                                                                 | 0.00                                  |
| 5.2.4.3: Efforts Made towards Housing for Local Communities                          |                                       |
| Number of Houses                                                                     | Number of Houses                      |
| 0                                                                                    | 0                                     |
| 5.2.4.4: Efforts Made towards Providing Transport to Local Communities               |                                       |
| Number of Beneficiaries                                                              | Number of Beneficiaries               |
| 0                                                                                    | 0                                     |
| 5.2.4.5: Efforts Made towards Providing Healthcare to Local Communities              |                                       |
| Number of Beneficiaries                                                              | Number of Beneficiaries               |
| 10                                                                                   | 45                                    |
| 5.2.4.6: Efforts Made towards Providing Hygiene & Sanitation to Local Communities    |                                       |
| Number of Beneficiaries                                                              | Number of Beneficiaries               |
| 10                                                                                   | 35                                    |

| 5.2.4.7: Efforts Made towards Skill Development Programs to Local Communities                  |                                                 |                                       |
|------------------------------------------------------------------------------------------------|-------------------------------------------------|---------------------------------------|
| Number of Beneficiaries                                                                        |                                                 | Number of Beneficiaries               |
| 10                                                                                             |                                                 | 45                                    |
| 5.2.4.8: Efforts Made to Promote Education & Knowledge Based Initiatives                       |                                                 |                                       |
| Number of Beneficiaries                                                                        |                                                 | Number of Beneficiaries               |
| 10                                                                                             |                                                 | 45                                    |
| 5.2.4.9: Communication Facilities Provided to Local Communities                                |                                                 |                                       |
| Number of Beneficiaries                                                                        |                                                 | Number of Beneficiaries               |
| 0                                                                                              |                                                 | 0                                     |
| 5.2.4.10: Any Other Steps Taken for Improving the Socio-Economic Standard of Local Communities |                                                 |                                       |
| Number of Beneficiaries                                                                        |                                                 | Number of Beneficiaries               |
| 10                                                                                             |                                                 | 50                                    |
| 5.2.4.11: Adoption of ODF                                                                      |                                                 |                                       |
| Number of Toilets Built inside the Lease Area                                                  | Number of Toilets Built outside the Lease Area: | Number of Beneficiaries               |
| 0                                                                                              | 0                                               | 0                                     |
| 5.2.4.12: Awareness Program among Mine Workers for Swatchata                                   |                                                 |                                       |
| Number of Swatchata Programmes Proposed                                                        |                                                 | Number of Swatchata Programmes Held   |
| 1                                                                                              |                                                 | 5                                     |
| 5.2.4.13: Efforts for green energy                                                             |                                                 |                                       |
| Total energy consumption (KWh)                                                                 |                                                 | Green energy consumption (% of total) |
| 50.00                                                                                          |                                                 | 40.00                                 |
| 5.2.4.14: Water & recycled use                                                                 |                                                 |                                       |
| Total water consumption (KLD)                                                                  |                                                 | Water recycled (% of total)           |
| 52.70                                                                                          |                                                 | 0.00                                  |

5.2.5: Year5

| Details of Work Proposed during the Year / Measures Planned for the Affected Segment | Cumulative Work done / Measures Taken |
|--------------------------------------------------------------------------------------|---------------------------------------|
| 5.2.5.1: Area to be Developed for Recreation                                         |                                       |
| Area (Ha)                                                                            | Area (Ha)                             |
| 0.00                                                                                 | 0.00                                  |
| 5.2.5.2: Area for Water Storage & Recharge Facility                                  |                                       |
| Area (Ha)                                                                            | Area (Ha)                             |
| 0.00                                                                                 | 0.00                                  |
| 5.2.5.3: Efforts Made towards Housing for Local Communities                          |                                       |
| Number of Houses                                                                     | Number of Houses                      |
| 0                                                                                    | 0                                     |
| 5.2.5.4: Efforts Made towards Providing Transport to Local Communities               |                                       |
| Number of Beneficiaries                                                              | Number of Beneficiaries               |
| 0                                                                                    | 0                                     |
| 5.2.5.5: Efforts Made towards Providing Healthcare to Local Communities              |                                       |
| Number of Beneficiaries                                                              | Number of Beneficiaries               |
| 10                                                                                   | 45                                    |
| 5.2.5.6: Efforts Made towards Providing Hygiene & Sanitation to Local Communities    |                                       |
| Number of Beneficiaries                                                              | Number of Beneficiaries               |
| 15                                                                                   | 40                                    |
| 5.2.5.7: Efforts Made towards Skill Development Programs to Local Communities        |                                       |
| Number of Beneficiaries                                                              | Number of Beneficiaries               |
| 10                                                                                   | 45                                    |

| 5.2.5.8: Efforts Made to Promote Education & Knowledge Based Initiatives                       |                                                 |                                       |
|------------------------------------------------------------------------------------------------|-------------------------------------------------|---------------------------------------|
| Number of Beneficiaries                                                                        |                                                 | Number of Beneficiaries               |
| 10                                                                                             |                                                 | 45                                    |
| 5.2.5.9: Communication Facilities Provided to Local Communities                                |                                                 |                                       |
| Number of Beneficiaries                                                                        |                                                 | Number of Beneficiaries               |
| 0                                                                                              |                                                 | 0                                     |
| 5.2.5.10: Any Other Steps Taken for Improving the Socio-Economic Standard of Local Communities |                                                 |                                       |
| Number of Beneficiaries                                                                        |                                                 | Number of Beneficiaries               |
| 10                                                                                             |                                                 | 50                                    |
| 5.2.5.11: Adoption of ODF                                                                      |                                                 |                                       |
| Number of Toilets Built inside the Lease Area                                                  | Number of Toilets Built outside the Lease Area: | Number of Beneficiaries               |
| 0                                                                                              | 0                                               | 0                                     |
| 5.2.5.12: Awareness Program among Mine Workers for Swatchata                                   |                                                 |                                       |
| Number of Swatchata Programmes Proposed                                                        |                                                 | Number of Swatchata Programmes Held   |
| 1                                                                                              |                                                 | 5                                     |
| 5.2.5.13: Efforts for green energy                                                             |                                                 |                                       |
| Total energy consumption (KWh)                                                                 |                                                 | Green energy consumption (% of total) |
| 50.00                                                                                          |                                                 | 50.00                                 |
| 5.2.5.14: Water & recycled use                                                                 |                                                 |                                       |
| Total water consumption (KLD)                                                                  |                                                 | Water recycled (% of total)           |
| 52.70                                                                                          |                                                 | 0.00                                  |

5.3: Rehabilitation & Resettlement of Affected Persons

|            |       |       |       |       |       |
|------------|-------|-------|-------|-------|-------|
| Particular | Year1 | Year2 | Year3 | Year4 | Year5 |
|------------|-------|-------|-------|-------|-------|



|                                                                                |    |   |   |   |   |
|--------------------------------------------------------------------------------|----|---|---|---|---|
| Proposed Number of Project Affected Persons(PAP)                               | 15 | 0 | 0 | 0 | 0 |
| Proposed Number of Person for Alternate Arrangement for Sustainable Livelihood | 15 | 0 | 0 | 0 | 0 |
| Proposed Number of Person for Skill Training                                   | 5  | 0 | 0 | 0 | 0 |
| Proposed Number of Person Likely to get Direct Employment                      | 3  | 0 | 0 | 0 | 0 |
| Proposed Number of Person Likely to get Indirect Employment                    | 2  | 0 | 0 | 0 | 0 |
| Proposed Project Affected Families Skilled and Absorbed                        | 3  | 0 | 0 | 0 | 0 |
| Proposed Number of Project Affected Families                                   | 3  | 0 | 0 | 0 | 0 |

Chapter 6: Progressive Mine Closure Plan

6.1: Status of Land

| Total Area Degraded                      |                             |                           |                                      |                                 | Total mined out area Reclaimed and Rehabilitated        |                                                                |                                                               | Other Areas Reclaimed and Rehabilitated         |                                        |
|------------------------------------------|-----------------------------|---------------------------|--------------------------------------|---------------------------------|---------------------------------------------------------|----------------------------------------------------------------|---------------------------------------------------------------|-------------------------------------------------|----------------------------------------|
| Total area under excavation in the lease |                             | Area under Dumps(in hect) | Area under utility services(in hect) | Area under Stack yards(in hect) | Mined out Area Reclaimed but not rehabilitated(in hect) | Mined outArea fully Rehabilitated from Reclaimed area(in hect) | Area under Water Reservoir considered Rehabilitated (in hect) | Stabililized Waste dump Rehabilitated (in hect) | Virgin area under Green Belt (in hect) |
| Area under mining operation              | Mined Out area in the lease |                           |                                      |                                 |                                                         |                                                                |                                                               |                                                 |                                        |
| 0.00                                     | 0.00                        | 0.00                      | 0.03                                 | 0.00                            | 0.00                                                    | 0.00                                                           | 0.00                                                          | 0.00                                            | 0.00                                   |

6.2: Progressive Reclamation and Rehabilitation Plan

6.2.1: Backfilling

|                                                          |          |
|----------------------------------------------------------|----------|
| Quantity of Waste / Fill Material Available at Site (m³) | 51120.00 |
| Availability of Top Soil for Spreading (m³)              | 13852.00 |
| Proposed Spread Area (m²)                                | 1267.00  |

6.2.1.1: Year Wise Proposal

| Sl.No | Year  | Pit ID | Area (m²) | Top RL | Bottom RL | Estimated Expenditure (₹ INR) |
|-------|-------|--------|-----------|--------|-----------|-------------------------------|
| 1     | Year1 | Pit 1  | 8888.90   | 312    | 311       | 266667.00                     |
| 2     | Year2 | Pit 1  | 2425.00   | 312    | 311       | 72750.00                      |
| 3     | Year3 | Pit 1  | 2539.00   | 312    | 311       | 76170.00                      |



6.2.2: Water Reservoir

|                                   |        |
|-----------------------------------|--------|
| Average Rainfall of The Area (mm) | 393.00 |
| Proposed Area under Water Storage | 1.7870 |

6.2.2.1: Preparations For Ground Water Recharging

|                           |                                    |
|---------------------------|------------------------------------|
| 6.2.2.1.1: Drilling Holes |                                    |
| Year                      | Proposed no of Holes to be Drilled |
| Year1                     | Nil                                |
| Year2                     | Nil                                |
| Year3                     | Nil                                |
| Year4                     | Nil                                |
| Year5                     | Nil                                |

|                                            |                            |
|--------------------------------------------|----------------------------|
| 6.2.2.1.2:Preparation of Course Gravel Bed |                            |
| Year                                       | Proposed Area of Bed (LxW) |
| Year1                                      | Nil                        |
| Year2                                      | Nil                        |
| Year3                                      | Nil                        |
| Year4                                      | Nil                        |
| Year5                                      | Nil                        |

|                           |
|---------------------------|
| Please specify, if others |
| Nil                       |

6.2.2.2: Protective measures (Please specify running meter)

|                    |
|--------------------|
| 6.2.2.2.1: Fencing |
|--------------------|

| Year  | Proposed Fencing Length (m) | Latitude(dd:mm:ss.ss) |             | Longitude(dd:mm:ss.ss) |             |
|-------|-----------------------------|-----------------------|-------------|------------------------|-------------|
|       |                             | From                  | To          | From                   | To          |
| Year1 | 193                         | 26:58:09.23           | 26:58:12.01 | 73:29:20.78            | 73:29:16.90 |
| Year2 | 22                          | 26:58:12.01           | 26:58:12.79 | 73:29:16.90            | 73:29:16.90 |
| Year3 | 23                          | 26:58:09.23           | 26:58:09.24 | 73:29:20.78            | 73:29:21.65 |
| Year4 | 243                         | 26:58:09.24           | 26:58:12.79 | 73:29:21.65            | 73:29:16.90 |
| Year5 | 0                           | Nil                   | Nil         | Nil                    | Nil         |

| 6.2.2.2: Retaining Wall |                          |                        |             |                         |             |
|-------------------------|--------------------------|------------------------|-------------|-------------------------|-------------|
| Year                    | Proposed Wall Length (m) | Latitude (dd:mm:ss.ss) |             | Longitude (dd:mm:ss.ss) |             |
|                         |                          | From                   | To          | From                    | To          |
| Year1                   | 218                      | 26:58:12.93            | 26:58:15.27 | 73:29:19.86             | 73:29:19.91 |
| Year2                   | 268                      | 26:58:15.28            | 26:58:12.91 | 73:29:24.36             | 73:29:24.40 |
| Year3                   | 0                        | Nil                    | Nil         | Nil                     | Nil         |
| Year4                   | 0                        | Nil                    | Nil         | Nil                     | Nil         |
| Year5                   | 0                        | Nil                    | Nil         | Nil                     | Nil         |

| 6.2.2.2.3: Garland Drains |                          |                        |             |                         |             |
|---------------------------|--------------------------|------------------------|-------------|-------------------------|-------------|
| Year                      | Proposed Bund Length (m) | Latitude (dd:mm:ss.ss) |             | Longitude (dd:mm:ss.ss) |             |
|                           |                          | From                   | To          | From                    | To          |
| Year1                     | 432                      | 26:58:09.22            | 26:58:15.31 | 73:29:20.77             | 73:29:19.63 |
| Year2                     | 266                      | 26:58:09.21            | 26:58:15.35 | 73:29:20.76             | 73:29:24.35 |
| Year3                     | 22                       | 26:58:15.35            | 26:58:09.21 | 73:29:24.35             | 73:29:21.64 |
| Year4                     | Nil                      | Nil                    | Nil         | Nil                     | Nil         |
| Year5                     | Nil                      | Nil                    | Nil         | Nil                     | Nil         |

6.2.3: Green Belt Development

6.2.3.1: Cumulative work done (upto end of previous block of five years)

| SI.No | Total Expenditure Incurred up to Last Year (INR) | Area Covered (Ha) | Number of Plants | Survival Rate (%) |
|-------|--------------------------------------------------|-------------------|------------------|-------------------|
| 1     | Nil                                              | Nil               | Nil              | Nil               |

6.2.3.2: Year Wise Proposal

| SI.No | Year  | Green Belt Location (s) | Area Proposed to be Covered (Ha) | Number of Plants Proposed | Expected Survival Rate (%) | Estimated Expenditure (₹ INR) |
|-------|-------|-------------------------|----------------------------------|---------------------------|----------------------------|-------------------------------|
| 1     | Year1 | Safety Barrier          | 0.0267                           | 316                       | 80                         | 316000                        |
| 2     | Year2 | Safety Barrier          | 0.0267                           | 316                       | 80                         | 316000                        |
| 3     | Year3 | Safety Barrier          | 0.0267                           | 316                       | 80                         | 316000                        |
| 4     | Year4 | Safety Barrier          | 0.0210                           | 316                       | 80                         | 316000                        |
| 5     | Year5 | Safety Barrier          | 0.0254                           | 316                       | 80                         | 316000                        |

6.2.4: Use of Shallow Pits

6.2.4.1: Cumulative Work Done (upto end of previous block of five years)

| SI.No | Pit ID | Work Done | Area covered (m <sup>2</sup> ) | Total Expenditure Incurred (up to last five year block) (₹ INR) |
|-------|--------|-----------|--------------------------------|-----------------------------------------------------------------|
| 1     | Nil    | Nil       | Nil                            | Nil                                                             |

6.2.4.2: Year Wise Proposal

| SI.No | Year | Pit ID | Total Area(Ha) | Area Proposed for Crops (Ha) | Suitable Crops | Area Proposed for Grass (Ha) | Total Proposed Expenditure | Latitude (dd:mm:ss.ss) | Longitude (dd:mm:ss.ss) | Remarks |
|-------|------|--------|----------------|------------------------------|----------------|------------------------------|----------------------------|------------------------|-------------------------|---------|
|-------|------|--------|----------------|------------------------------|----------------|------------------------------|----------------------------|------------------------|-------------------------|---------|

|   |     |     |     |     |     |     |         |      |     |      |     |     |
|---|-----|-----|-----|-----|-----|-----|---------|------|-----|------|-----|-----|
|   |     |     |     |     |     |     | (₹ INR) | From | To  | From | To  |     |
| 1 | Nil | Nil | Nil | Nil | Nil | Nil | Nil     | Nil  | Nil | Nil  | Nil | Nil |

6.2.5: Pisciculture

|                                                      |     |
|------------------------------------------------------|-----|
| 6.2.5.1: Total Expenditure incurred as on Date (INR) | Nil |
|------------------------------------------------------|-----|

6.2.5.2: Cumulative work done as on Date

|       |        |                        |                     |
|-------|--------|------------------------|---------------------|
| SI.No | Pit ID | Area (m <sup>2</sup> ) | Expenditure (₹ INR) |
| 1     | Nil    | Nil                    | Nil                 |

6.2.5.3: Year Wise Proposal

|       |      |        |                        |                               |
|-------|------|--------|------------------------|-------------------------------|
| SI.No | Year | Pit ID | Area (m <sup>2</sup> ) | Estimated Expenditure (₹ INR) |
| 1     | Nil  | Nil    | Nil                    | Nil                           |

|                                           |     |
|-------------------------------------------|-----|
| 6.2.5.4: Source of Water for Pisciculture | Nil |
|-------------------------------------------|-----|

|                                                                                                 |     |
|-------------------------------------------------------------------------------------------------|-----|
| 6.2.5.5: Whether the quality of water has been assessed & found to be suitable for Pisciculture | Yes |
|-------------------------------------------------------------------------------------------------|-----|

6.2.6: Recreational Facility

|                                                                        |     |
|------------------------------------------------------------------------|-----|
| 6.2.6.1: Total Expenditure Incurred (up to last five year block) (INR) | Nil |
|------------------------------------------------------------------------|-----|

6.2.6.2: Cumulative work done as on Date

|       |        |                        |                     |
|-------|--------|------------------------|---------------------|
| SI.No | Pit ID | Area (m <sup>2</sup> ) | Expenditure (₹ INR) |
| 1     | Nil    | Nil                    | Nil                 |

6.2.6.3: Year Wise Proposal

| SI.No | Year  | Type of Recreational Facility | Area Covered (Ha) | Latitude (dd:mm:ss.ss) |     | Longitude (dd:mm:ss.ss) |     | Estimated Expenditure (INR) |
|-------|-------|-------------------------------|-------------------|------------------------|-----|-------------------------|-----|-----------------------------|
|       |       |                               |                   | From                   | To  | From                    | To  |                             |
| 1     | Year1 | Nil                           | Nil               | Nil                    | Nil | Nil                     | Nil | Nil                         |
| 2     | Year2 | Nil                           | Nil               | Nil                    | Nil | Nil                     | Nil | Nil                         |
| 3     | Year3 | Nil                           | Nil               | Nil                    | Nil | Nil                     | Nil | Nil                         |
| 4     | Year4 | Nil                           | Nil               | Nil                    | Nil | Nil                     | Nil | Nil                         |
| 5     | Year5 | Nil                           | Nil               | Nil                    | Nil | Nil                     | Nil | Nil                         |

6.2.7: Dump Area Stabilization & Development

| SI.No | Year | Dump ID | No of Terraces | Average Height of Terraces (m) | Length of Toe Wall (m) | Length of Garland Drain (m) | Area Stabilized (Ha) | Method of Stabilization | Estimated Expenditure (₹ INR) | No of Check Dams |
|-------|------|---------|----------------|--------------------------------|------------------------|-----------------------------|----------------------|-------------------------|-------------------------------|------------------|
| 1     | Nil  | Nil     | Nil            | Nil                            | Nil                    | Nil                         | Nil                  | Nil                     | Nil                           | Nil              |

6.2.8: Other Form of Reclaiming the Area

6.2.8.1: Cumulative work done as on Date

| SI.No | Total Expenditure incurred as on Date (INR) | Work Done |
|-------|---------------------------------------------|-----------|
| 1     | Nil                                         | Nil       |

6.2.8.2: Year Wise Proposal

| SI.No | Year  | Work Proposals | Estimated Expenditure (INR) |
|-------|-------|----------------|-----------------------------|
| 1     | Year1 | Nil            | Nil                         |
| 2     | Year2 | Nil            | Nil                         |
|       |       |                |                             |

|   |       |     |     |
|---|-------|-----|-----|
| 3 | Year3 | Nil | Nil |
| 4 | Year4 | Nil | Nil |
| 5 | Year5 | Nil | Nil |

6.2.9: TopSoil Management

6.2.9.1: Cumulative Work Done as on Date

| SI.No | Top Soil Generated (m³) | Top Soil Utilized (m³) | Topsoil Stored (m³) | Total expenditure incurred as on date (₹) |
|-------|-------------------------|------------------------|---------------------|-------------------------------------------|
| 1     | Nil                     | Nil                    | Nil                 | Nil                                       |

6.2.9.2: Year Wise Proposal

| SI.No | Year  | Topsoil Generated (m³) (A) | Topsoil Utilized (m³) (B) | Topsoil Stored (m³) (A-B) | Estimated Expenditure (INR) |
|-------|-------|----------------------------|---------------------------|---------------------------|-----------------------------|
| 1     | Year1 | 8889.00                    | 2770.00                   | 6119.00                   | 191630.00                   |
| 2     | Year2 | 2425.00                    | 2770.00                   | -345.00                   | 62350.00                    |
| 3     | Year3 | 2539.00                    | 2770.00                   | -231.00                   | 64600.00                    |
| 4     | Year4 | 0.00                       | 2770.00                   | -2770.00                  | 13850.00                    |
| 5     | Year5 | 0.00                       | 2770.00                   | -2770.00                  | 13850.00                    |

6.2.10: Tailings Dam Management

| SI.No | Year  | Yearly generation of Tailing (m³) (A) | Total capacity of Tailing Pond (m³) | Measures Proposed for Periodic Desilting | Yearly Utilization of Tailing (m³) (B) | Disposal of Tailing to Tailing Pond (m³) (A-B) | Tailing Dam Design | Structural Stability Studies |
|-------|-------|---------------------------------------|-------------------------------------|------------------------------------------|----------------------------------------|------------------------------------------------|--------------------|------------------------------|
| 1     | Year1 | Nil                                   | Nil                                 | Nil                                      | Nil                                    | Nil                                            | Nil                | Nil                          |
| 2     | Year2 | Nil                                   | Nil                                 | Nil                                      | Nil                                    | Nil                                            | Nil                | Nil                          |



|   |       |     |     |     |     |     |     |     |
|---|-------|-----|-----|-----|-----|-----|-----|-----|
| 3 | Year3 | Nil | Nil | Nil | Nil | Nil | Nil | Nil |
| 4 | Year4 | Nil | Nil | Nil | Nil | Nil | Nil | Nil |
| 5 | Year5 | Nil | Nil | Nil | Nil | Nil | Nil | Nil |

6.2.11: Land Use of Lease Area at the Expiry of Lease Period

| Total Area Degraded         |                           |                            |                                      | Non Degraded area       | Total mined out area Reclaimed and Rehabilitated          |                                                                |                                                               | Other Areas Reclaimed and Rehabilitated         |                                        |                                                    |                                                |
|-----------------------------|---------------------------|----------------------------|--------------------------------------|-------------------------|-----------------------------------------------------------|----------------------------------------------------------------|---------------------------------------------------------------|-------------------------------------------------|----------------------------------------|----------------------------------------------------|------------------------------------------------|
| Mined Out area in the lease | Area under Dumps(in hect) | Area under the Tailing Dam | Area under utility services(in hect) | Area undisturbed/virgin | Mined out Area Reclaimed but not rehabilitated(in n hect) | Mined outArea fully Rehabilitated from Reclaimed area(in hect) | Area under Water Reservoir considered Rehabilitated (in hect) | Stabililized Waste dump Rehabilitated (in hect) | Virgin area under Green Belt (in hect) | Rehabilitated Area under utility services(in hect) | Rehabilitated Area under Tailing dam (in hect) |
| 4.16                        | 0.00                      | 0.00                       | 0.01                                 | 0.63                    | 0.00                                                      | 0.80                                                           | 3.36                                                          | 0.00                                            | 0.63                                   | 0.01                                               | 0.00                                           |

Chapter 7: Financial Assurance/ Performance Surety (AREA PUT TO USE)

Year1

Consolidated View of Financial Assurance

| SI.No | Particular                                              | Area put to use at Start of Year (ha)<br>(A) | Additional Requirement (ha) (B) | Total (ha) (C = A + B) |
|-------|---------------------------------------------------------|----------------------------------------------|---------------------------------|------------------------|
| 1     | Area under Mining                                       | 0.00                                         | 0.89                            | 0.89                   |
| 2     | Topsoil stacking                                        | 0.00                                         | 0.17                            | 0.17                   |
| 3     | Overburden/Waste Dumping                                | 0.00                                         | 0.50                            | 0.50                   |
| 4     | Mineral Storage                                         | 0.00                                         | 0.01                            | 0.01                   |
| 5     | Infrastructure (Workshop, Administrative Building etc.) | 0.01                                         | 0.01                            | 0.02                   |
| 6     | Roads                                                   | 0.04                                         | 0.06                            | 0.10                   |
| 7     | Railway                                                 | 0.00                                         | 0.00                            | 0.00                   |
| 8     | Tailing Pond                                            | 0.00                                         | 0.00                            | 0.00                   |
| 9     | Effluent Treatment Plant                                | 0.00                                         | 0.00                            | 0.00                   |
| 10    | Mineral Separation Plant                                | 0.00                                         | 0.00                            | 0.00                   |
| 11    | Township Area                                           | 0.00                                         | 0.00                            | 0.00                   |
| 12    | Others to specify                                       | 0.00                                         | 0.00                            | 0.00                   |
|       | Total                                                   | 0.05                                         | 1.64                            | 1.69                   |

Year2

Consolidated View of Financial Assurance



| SI.No | Particular                                                 | Area put to use at Start of Year (ha)<br>(A) | Additional Requirement (ha) (B) | Total (ha) (C = A + B) |
|-------|------------------------------------------------------------|----------------------------------------------|---------------------------------|------------------------|
| 1     | Area under Mining                                          | 0.89                                         | 0.24                            | 1.13                   |
| 2     | Topsoil stacking                                           | 0.17                                         | 0.04                            | 0.21                   |
| 3     | Overburden/Waste Dumping                                   | 0.50                                         | 0.98                            | 1.48                   |
| 4     | Mineral Storage                                            | 0.01                                         | 0.00                            | 0.01                   |
| 5     | Infrastructure (Workshop,<br>Administrative Building etc.) | 0.02                                         | 0.00                            | 0.02                   |
| 6     | Roads                                                      | 0.10                                         | 0.00                            | 0.10                   |
| 7     | Railway                                                    | 0.00                                         | 0.00                            | 0.00                   |
| 8     | Tailing Pond                                               | 0.00                                         | 0.00                            | 0.00                   |
| 9     | Effluent Treatment Plant                                   | 0.00                                         | 0.00                            | 0.00                   |
| 10    | Mineral Separation Plant                                   | 0.00                                         | 0.00                            | 0.00                   |
| 11    | Township Area                                              | 0.00                                         | 0.00                            | 0.00                   |
| 12    | Others to specify                                          | 0.00                                         | 0.00                            | 0.00                   |
|       | Total                                                      | 1.69                                         | 1.26                            | 2.95                   |

**Year3**

## Consolidated View of Financial Assurance

| SI.No | Particular                                                 | Area put to use at Start of Year (ha)<br>(A) | Additional Requirement (ha) (B) | Total (ha) (C = A + B) |
|-------|------------------------------------------------------------|----------------------------------------------|---------------------------------|------------------------|
| 1     | Area under Mining                                          | 1.13                                         | 0.26                            | 1.39                   |
| 2     | Topsoil stacking                                           | 0.21                                         | 0.07                            | 0.28                   |
| 3     | Overburden/Waste Dumping                                   | 1.48                                         | 0.00                            | 1.48                   |
| 4     | Mineral Storage                                            | 0.01                                         | 0.00                            | 0.01                   |
| 5     | Infrastructure (Workshop,<br>Administrative Building etc.) | 0.02                                         | 0.00                            | 0.02                   |

|    |                          |      |      |      |
|----|--------------------------|------|------|------|
| 6  | Roads                    | 0.10 | 0.00 | 0.10 |
| 7  | Railway                  | 0.00 | 0.00 | 0.00 |
| 8  | Tailing Pond             | 0.00 | 0.00 | 0.00 |
| 9  | Effluent Treatment Plant | 0.00 | 0.00 | 0.00 |
| 10 | Mineral Separation Plant | 0.00 | 0.00 | 0.00 |
| 11 | Township Area            | 0.00 | 0.00 | 0.00 |
| 12 | Others to specify        | 0.00 | 0.00 | 0.00 |
|    | Total                    | 2.95 | 0.33 | 3.28 |

Year4

Consolidated View of Financial Assurance

| Sl.No | Particular                                              | Area put to use at Start of Year (ha)<br>(A) | Additional Requirement (ha) (B) | Total (ha) (C = A + B) |
|-------|---------------------------------------------------------|----------------------------------------------|---------------------------------|------------------------|
| 1     | Area under Mining                                       | 1.39                                         | 0.00                            | 1.39                   |
| 2     | Topsoil stacking                                        | 0.28                                         | 0.00                            | 0.28                   |
| 3     | Overburden/Waste Dumping                                | 1.48                                         | 0.00                            | 1.48                   |
| 4     | Mineral Storage                                         | 0.01                                         | 0.00                            | 0.01                   |
| 5     | Infrastructure (Workshop, Administrative Building etc.) | 0.02                                         | 0.00                            | 0.02                   |
| 6     | Roads                                                   | 0.10                                         | 0.00                            | 0.10                   |
| 7     | Railway                                                 | 0.00                                         | 0.00                            | 0.00                   |
| 8     | Tailing Pond                                            | 0.00                                         | 0.00                            | 0.00                   |
| 9     | Effluent Treatment Plant                                | 0.00                                         | 0.00                            | 0.00                   |
| 10    | Mineral Separation Plant                                | 0.00                                         | 0.00                            | 0.00                   |
| 11    | Township Area                                           | 0.00                                         | 0.00                            | 0.00                   |
| 12    | Others to specify                                       | 0.00                                         | 0.00                            | 0.00                   |
|       | Total                                                   | 3.28                                         | 0.00                            | 3.28                   |

Year5

Consolidated View of Financial Assurance

| SI.No | Particular                                              | Area put to use at Start of Year (ha)<br>(A) | Additional Requirement (ha) (B) | Total (ha) (C = A + B) |
|-------|---------------------------------------------------------|----------------------------------------------|---------------------------------|------------------------|
| 1     | Area under Mining                                       | 1.39                                         | 0.00                            | 1.39                   |
| 2     | Topsoil stacking                                        | 0.28                                         | 0.00                            | 0.28                   |
| 3     | Overburden/Waste Dumping                                | 1.48                                         | 0.00                            | 1.48                   |
| 4     | Mineral Storage                                         | 0.01                                         | 0.00                            | 0.01                   |
| 5     | Infrastructure (Workshop, Administrative Building etc.) | 0.02                                         | 0.00                            | 0.02                   |
| 6     | Roads                                                   | 0.10                                         | 0.00                            | 0.10                   |
| 7     | Railway                                                 | 0.00                                         | 0.00                            | 0.00                   |
| 8     | Tailing Pond                                            | 0.00                                         | 0.00                            | 0.00                   |
| 9     | Effluent Treatment Plant                                | 0.00                                         | 0.00                            | 0.00                   |
| 10    | Mineral Separation Plant                                | 0.00                                         | 0.00                            | 0.00                   |
| 11    | Township Area                                           | 0.00                                         | 0.00                            | 0.00                   |
| 12    | Others to specify                                       | 0.00                                         | 0.00                            | 0.00                   |
|       | Total                                                   | 3.28                                         | 0.00                            | 3.28                   |
|       | Grand Total                                             |                                              |                                 | 3.28                   |

|                    |
|--------------------|
| Performance Surety |
|--------------------|

(b) Performance Surety

| Lease Category (A/B) | Total Resources in Tonnes For Calculations of Performnce Surety | Existing Performnce Surety in Rs | Valid Till (dd/mm/yyyy) | Upload Copy Of Existing Performance Surety as Attachment |
|----------------------|-----------------------------------------------------------------|----------------------------------|-------------------------|----------------------------------------------------------|
| A                    | 1500000                                                         | 3804375.00                       | 02/05/2030              | <a href="#">BG-Block_18_(1).pdf</a>                      |

Approved

# **Chapter 8 : Review of previous proposals during the last plan period (NA)**

Approved

Chapter 9: Impact Assessment(for fresh grant)

9.1 : Baseline Information

|                                                        |    |
|--------------------------------------------------------|----|
| Whether Area falls under Forest                        | No |
| Whether Area falls under Wildlife Sanctuary            | No |
| Whether Area falls under Coastal Regulation Zone (CRZ) | No |
| Whether Area falls under Defence Land                  | No |
| Any Other Clearance (specify)                          | No |

Any Significant Objections from any Agency Involved in Stakeholder’s Consultation.

No

(Total 500 characters)

9.2: Environment Parameters

9.2.1: Environment Monitoring

Monitoring Activity

9.2.1.1: Ambient Air Quality

|                                          |     |
|------------------------------------------|-----|
| Core Zone (Quarterly Monitoring Planned) | Yes |
|------------------------------------------|-----|



|                                            |     |
|--------------------------------------------|-----|
| Buffer Zone (Quarterly Monitoring Planned) | Yes |
|--------------------------------------------|-----|

9.2.1.2: Water Quality

|                                            |     |
|--------------------------------------------|-----|
| Core Zone (Quarterly Monitoring Planned)   | Yes |
| Buffer Zone (Quarterly Monitoring Planned) | Yes |

9.2.1.3: Noise Level

|                                            |     |
|--------------------------------------------|-----|
| Core Zone (Quarterly Monitoring Planned)   | Yes |
| Buffer Zone (Quarterly Monitoring Planned) | Yes |

9.3: Impact Assessment

9.3.2: Land Environment

9.3.2.1: BASE / PRESENT STATUS

| Pre Mining Use                                                                                                     | AREA (Ha) |
|--------------------------------------------------------------------------------------------------------------------|-----------|
| Barren / Waste land with small bushes & shrubs                                                                     | 2.6456    |
| Land under Agriculture / Crops                                                                                     | 2.1544    |
| Land covered with Plants                                                                                           | 0.0000    |
| Land under Grass Cover                                                                                             | 0.0000    |
| Land under Public Infrastructure / Utilities (water bodies, roads, railways, electric lines, telephone lines etc.) | 0.0000    |

|                                                                    |            |
|--------------------------------------------------------------------|------------|
| Land under Habitation                                              | 0.0312     |
| Land under Monuments & places of Historical Importance             | 0.0000     |
| Degraded by Pits & Excavation                                      | 0.0000     |
| Degraded by Dumps & Material Staking                               | 0.0000     |
| Covered under Mine Infrastructure (plants, shades, buildings etc.) | 0.0000     |
| Land under Forest                                                  | 0.0000     |
| Historically, Culturally & Ecologically Important Places           | 0.0000     |
| Any Other, please specify below                                    | 0.0000     |
| Date of Observation                                                | 10/03/2025 |

9.3.2.2: ANTICIPATED IMPACT

| Post Mining Use                          | AREA (Ha) |
|------------------------------------------|-----------|
| Degradation by Excavation                | 1.3958    |
| Degradation by Dumps & Material Staking  | 1.4549    |
| Covered under Plants, Shades & Buildings | 0.6376    |
| Covered by Roads & Approaches            | 0.0274    |
| Any Other, please specify below          | 0.0000    |

9.3.2.3: MITIGATION MEASURES

9.3.2.3.1: Backfilling



Land degradation is one of the major adverse impacts of open-cast mining activities and any effort to control adverse impacts would be incomplete without appropriate land reclamation strategy. After the exhaustion of entire mineable limestone ,part of mined out pit will be backfilled by overburden and waste and remaining mined out pit will be converted into a water reservoir for utilization for irrigation purpose. The water reservoir will be properly fenced by barbed wire or a bund will be constructed outside the water reservoir during the closure of the mine. At the end of conceptual total 0.80 hect. Area will be backfilled after complete extraction of mineral.

(Total 1000 characters)

**9.3.2.3.2: Area proposed to be covered by Plantation in Backfilled Area**

During five year plan period backfilling has not proposed in excavated pit. At the conceptual stage an area of 0.80 ha shall be backfilled and plantation & re-grassing will be done over the backfilled area.

(Total 1000 characters)

**9.3.2.3.3: Proposed Area under Agriculture**

Not proposed

(Total 1000 characters)

**9.3.2.3.4: Proposed Area to be converted to Grazing Land**

Not proposed

(Total 1000 characters)

**9.3.2.3.5: Ground Water Recharging**

The excavated pit (Area-3.36 hect.) will be a ground water recharge.

(Total 1000 characters)

**9.3.2.3.6: Green Belt Development**

•In the next five year plan period total plantation will be carried out over an area0.1267 hect in7.5m safety barrier area of block area. The rate of plantation is1000 nos.of sapling per hectare. Plantation namely Mango, Pipal, Khejari, and Neem etc will be carried out in the area. • At the conceptual stage total greenbelt will cover 33% of the broken area which is 1.58 Ha. At the conceptual stage total 1580 sapling will be plantedoveranarea1.58hect. •The mining lease holder ,re-grassing will be carried out a long with plantation and restore the land to a condition which is fit for growth of flora & fauna.

(Total 1000 characters)

**9.3.2.3.7: Agriculture**

Not proposed

(Total 1000 characters)

9.3.3: Air Environment

9.3.3.1: Climate & Meteorology (Please provide average of 10 years)

|                       |       |
|-----------------------|-------|
| Temperature (°C)      |       |
| Maximum               | 45.00 |
| Minimum               | 1.00  |
| Relative Humidity (%) |       |
| 35.00                 |       |
| Average Rainfall (mm) |       |
| 393.00                |       |

9.3.3.2: Air Quality Details for Base line Information / Present Status

| S.N. | Station Name | Season | PM10 (µg/m3) | PM10 Excess (µg/m3)2 | PM2.5 (µg/m3) | PM2.5 Excess (µg/m3)2 | SO2 Value (µg/m3) | SO2 Excess (µg/m3) | NOX Value (µg/m3) | NOX Excess (µg/m3) | Date of Observation | Action |
|------|--------------|--------|--------------|----------------------|---------------|-----------------------|-------------------|--------------------|-------------------|--------------------|---------------------|--------|
| 1    | AAQ/08       | Winter | 88.5400      | 0.0000               | 46.2100       | 0.0000                | 9.7800            | 0.0000             | 21.6000           | 0.0000             | 08/02/2025          | Nil    |

9.3.3.3: Impact Assessment & Mitigation Measures

9.3.3.3.1: Anticipated Impact

Give details on Prediction of fugitive dust emissions due to mining activities, crushing & cleaning plants, loading & unloading, transportation by rail, road or conveyor

Fugitive dust emissions from mining activities pose a significant environmental and health concern, especially when these operations are situated near human settlements. These emissions originate from unconfined sources during drilling, blasting, excavation, hauling, and from wind erosion of exposed materials. Factors influencing dust generation include material moisture content, wind speed, frequency of operations, and the control measures in place. In nearby habitations, fine particulate matter (PM10 and PM2.5) can degrade air quality, leading to respiratory and cardiovascular issues, particularly among vulnerable groups such as children and the elderly. Dust deposition also impacts agriculture, contaminates water sources, and damages property. Miners face direct exposure to high levels of dust, which can result in silicosis, pneumoconiosis, skin and eye irritation, and long-term respiratory illnesses if proper protective measures are not taken.

(Total 1000 characters)

9.3.3.3.2: Mitigation Measure

Give details on measures to reduce the emissions of pollutants during mining, loading, unloading, transportation, drilling, blasting, crushing etc. to maintain the air quality

1)Water sprinkling will be done thrice during the day in summer season and in winter season for settling the dust particles on haul road, faces, crushing area and loading point etc. 2) Sharp drill bits will be used for drilling and they will be maintained periodically to reduce the generation of dust. 3) Transportation of mineral will be done on Kaccha road which will be metallized. 4) Drilling machines will have bag filters attached to them to prevent the dust to get air borne supplemented with wet drilling. 5) Blasting will be done in most scientific manner with the use of latest technology and optimizing the blasting parameters to control & prevent the dust to get air borne, and also to control the fly rocks.6) Regular maintenance of machinery and vehicles will be done to check the excess emissions. A system of regular over hauling of dumpers & excavators, after specified hours of working shall be developed and observed to avoid generation of obnoxious fumes. 7) Greenbelt with tall trees will be maintained which will restrict the particulates and reduce the concentration of SO2 and NO2.8) Plantation proposed along haul road and statutory barrier will also protect the soil from wind erosions.9) Protective gear will be provided to the workers for safe working.10)All the haulage roads will be kept properly maintained and watered regularly during the working shift to prevent generation of dust due to the movement of dumpers, water tankers etc. 11) Dust mask shall be provided to the workers engaged at dust generation points like dumping and loading points. 12) All heavy machinery operated by workmen, such as drill rigs, dozers, and haul trucks, should be equipped with closed cabins.

(Total 1000 characters)

9.3.4: Water Environment

9.3.4.1: RAIN WATER

9.3.4.1.1: Base / Present Status

(Details of Rivers, Springs, Lakes, Reservoirs & Drains up to First Order in Study Area)

There is no water body within the lease area

(Total 1000 characters)

9.3.4.1.2: Anticipated Impact

(Impact on Surface Water Bodies / Groundwater Table Regime / Streams / Lake / Springs due to Mining, to be Assessed from Hydro-geological StudyGive details about impact on vegetation)

There is no surface water body in the block area so there will be no impact on surface water bodies The mining shall not intercept the ground water table so there will be no impact on ground water The rain water accumulated in the pits will be used by local villagers for farming and other practices

(Total 1000 characters)

9.3.4.1.3: Mitigation Measure

(Possibilities of Rain Water Harvesting & Artificial Recharge with in the Mining Lease)

The mine out pit will be a water harvesting structure Rain water harvesting structure shall be built in future

(Total 1000 characters)

9.3.4.2: WATER BODY

9.3.4.2.1: Base / Present Status

(Water Bodies Existing & Water Bodies likely to be created due to Mining Activities & their Water Holding Capacity)

At present there are no existing water bodies within the mining area.

(Total 1000 characters)

9.3.4.2.2: Anticipated Impact

(Ingress of Sea Water, Particularly for Mining Projects in Coastal Areas)

No impact as the mentioned water body is away from block area.

(Total 1000 characters)

9.3.4.2.3: Mitigation Measure

(Steps to Minimize Impact on Water Table if Mining Intercepts Groundwater Regime)

Not Required due to above stated reason

(Total 1000 characters)

9.3.4.3: Water Balance

9.3.4.3.1: Base / Present Status

(Water Balance (Withdrawal of Surface Water & Release of Mine Drainage Water) Water Requirement & Waste Water Generation from various Activities of Mine, Including Beneficiation)

The accumulated rain water will be utilized for mining and other activities The water will be used inactivities like Drinking Domestic Use Green Belt Wet Drilling and Dust Suppression etc On yearly basis the water requirement will be 31 KLD for first year 41 KLD for second year 51 KLD fo rthird year 52 KLD for fourth year and 52 KLD for fifth year Source of supply Drinking water will be procured from government supply by water tanker from bairas village water for rest of the activities will be arranged from existing old Mines situated near Bairas village Disposal of water No water shall be discharge from mine area the water accumulated in the mine pit shall be used in water sprinkling wet drilling green belt development the domestic water shall be utilized for plantation purposes Extent of recycling the wet drilling water shall be recycling and reused for drilling purposes

(Total 1000 characters)

9.3.4.3.2: Anticipated Impact

(Impact of Water Drawl on Surface & Groundwater Resources Impact on Surface & Groundwater Quality due to Discharges from Mining, Tailings Pond, Workshop, Township, & Leach ate from Solid Waste Dumps etc)

There will be no impact on surface ground water as ther is no discharge from mining lease workshop leach ate from solid waste dump Waste Water Generation from office canteen workshop sanitations facilities will be utilized after treatment and plantation and water sprinkling

(Total 1000 characters)

9.3.4.3.3: Mitigation Measure

(Construction of Check Dams, Sedimentation Ponds, Settling Tanks, Retaining Walls etc. with Design & Site Features for Control of run-off Mine Water Treatment for Meeting the Prescribed Standard Waste Water Treatment for Township Sewage, Workshop(s), Tailing Pond Overflow etc)

Retaining walls will be strategically built around overburden dumps stockpiles and vulnerable slopes to prevent soil erosion and uncontrolled runoff These structures will stabilize the terrain and help direct runoff towards designated collection points aiding in effective water management Settling tanks will be constructed at key runoff collection points to capture suspended solids from mine water before it is reused or discharged These tanks will be designed to accommodate peak runoff volumes allowing sufficient retention time for sedimentation thereby improving water quality and reducing siltation downstream

(Total 1000 characters)

9.3.5: Noise

9.3.5.1: Critical Locations Identified within Lease Area

Noise is defined as an unwanted sound. The general Noise level remains within the permissible limit (80dBA) in the block area. At some places due to the operation of HEMM it increases beyond the permissible limit. The location are loading faces, drilling operations, while operating HEMM- 90 to95dBA, near Drilling operation- 92 to 98 dBA, Dumper cabin- 70 to 75dB Aand during Blasting- 105 to 125dBA.

(Total 1000 characters)

9.3.5.2: Give Detail About Prediction of Noise Level by using Mathematical Modeling at Different Locations Identified

ISO 9613-2 used for modeling 1. Source Characterization: Identify and characterize the noise sources in the mining environment, such as heavy machinery (drills, haulage trucks), transportation vehicles, and blasting operations. 2. Geometry and Material Modeling: Accurately model the geometry of the mining site, including the location of noise sources, surrounding terrain, and potential barriers or reflectors. 3. Grid and Receptor Placement: Define a grid of points (receptors) at which noise levels need to be predicted. 4. ISO 9613-2 Calculations: Use the ISO 9613-2 methodology to calculate sound pressure levels at each receptor point. 5. Noise Mapping: Represent the predicted noise levels at each receptor point on a map, creating a noise contour map. Near operation of HEMM- 90 to95dBA Near Drilling operation- 92 to98dBA Dumper cabin- 70 to75dBA during Blasting- 105 to 115dBA

(Total 1000 characters)

9.3.5.3: Measures to Minimize the Impact on Receiving Environment

1)Proper maintenance, oiling and greasing of machines at regular intervals will be done to reduce noise generation. 2)Confining the noise generating sources. 3) In order to reduce the effect of noise pollution, ear plugs & ear muffs will be provided to all employees 4) Control blasting will be done.5) Plantation will be done.6) Periodic maintenance of machinery will be done to reduce noise level. 7) EV trucks in due course will be used.

(Total 1000 characters)

9.3.5.4: Noise Details for Base / Present Status

| Noise Standards |                  |                    |            |
|-----------------|------------------|--------------------|------------|
| Area Code       | Category of Area | Limits in dB(A)Leq |            |
|                 |                  | Day Time           | Night Time |
| A               | Industrial Area  | 75                 | 70         |
| B               | Commerical Area  | 65                 | 55         |
| C               | Residential Area | 55                 | 45         |
| D               | Silence Area     | 40                 | 50         |

Note :

1. Day time reckoned in between 6.00 am to 9.00p.m



- 2. Night time reckoned in between 9.00p.m.to 6:00am
- 3. Silence zone is defined as areas up to 100 meter around such premises as Hospitals, Educational institutes and Courts. The Silence zones are to be declared by the competent Authority
- 4. Mixed categories of areas should be declared as "one of the four above mentioned categories" by the Competent Authority and the corresponding standards shall be applied.

| S.N. | Station Name | Season | Type of Area | Noise At Day Time | Excess Noise At Day | Noise At Night Time | Excess Noise at Night | Date of Observation |
|------|--------------|--------|--------------|-------------------|---------------------|---------------------|-----------------------|---------------------|
| 1    | AN-040225-02 | Winter | Industrial   | 57.8              | 0                   | 57.8                | 0                     | 03/02/2025          |
| 2    | AN-040225-02 | Winter | Residential  | 43.5              | 0                   | 43.5                | 0                     | 03/02/2025          |

9.3.5.5: Impact Assessment & Mitigation Measures

9.3.5.5.1: Anticipated Impact

Give details on impact on ambient noise level due to rock excavation, transportation, processing equipment’s & ancillaries

Mining activities impact ambient noise levels differently based on the specific operation Drilling generates continuous high noise typically between 90 to 98 dBA due to the use of heavy rotary drills and pneumatic tools which can cause hearing fatigue for nearby workers Blasting produces short but intense noise spikes often reaching 105 to 110 dBA resulting in sudden disturbances to workers nearby residents and wildlife The operation of Heavy Earth Moving Machinery HEMM such as dumpers excavators and loaders typically produces noise in the range of 90 to 95 dBA creating continuous noise pollution at the mining site Transportation of materials by trucks and dumpers along haul roads adds additional noise generally between 85 and 95 dBA which can extend to surrounding areas depending on terrain and weather conditions Finally ancillary activities like workshops and maintenance generate intermittent noise usually between 70 and 85 dBA further adding to the overall noise environment Together these activities elevate ambient noise levels significantly necessitating effective noise control measures to protect worker health and minimize impact on nearby communities

(Total 1000 characters)

9.3.5.5.2: Mitigation Measure

Give details on measures for noise abatement including point source & line source

Noise abatement in mining operations requires targeted measures addressing both point sources and line sources of noise to effectively reduce ambient noise levels Point sources such as drilling rigs and blasting sites generate concentrated noise from specific locations To mitigate noise from these installing acoustic enclosures and barriers around drilling machines can significantly reduce sound emission For blasting employing controlled blasting techniques with delay detonators minimizes impulsive noise Regular maintenance of equipment ensures smoother operation with less noise while scheduling highnoise activities during daytime reduces community disturbance Line sources primarily represented by moving vehicles such as dumpers loaders and trucks on haul roads produce noise over extended areas as they travel Noise from line sources can be abated by maintaining and paving haul roads to minimize noise caused by uneven surfaces enforcing speed limits to reduce engine and tire noise and using vehicle mufflers and silencers Additionally establishing green belts or vegetative buffers along haul roads acts as a natural sound barriers absorbing

and scattering noise before it reaches sensitive receptors Providing operators with closed soundproof cabins helps protect workers from continuous noise exposure during machinery operation

(Total 1000 characters)

9.3.6: VIBRATION

9.3.6.1: Vibration Details for Base / Present Status

| S.N. | Station Name | Season | Distance from the<br>Blasting Site (m) | Peak Particle Velocity<br>(mm/s) | Air Over Pressure<br>(DB) | Frequency (Hz) | Date of Observation |
|------|--------------|--------|----------------------------------------|----------------------------------|---------------------------|----------------|---------------------|
| 1    | Nil          | Nil    | Nil                                    | Nil                              | Nil                       | Nil            | Nil                 |

9.3.6.2: Impact Assessment & Mitigation Measures

9.3.6.2.1: Anticipated Impact

(Give details on impact of vibrations including damage to materials/structures due to blasting)

Due to increase in vibration level the anticipated impact are 1 Due to vibration the machinery life reduces 2 The temporary structure mud structure develop cracks 3 The existing cracks widen

(Total 1000 characters)

9.3.6.2.2: Mitigation Measure

Give details on measures for noise abatement including point source & line source

Controlled blasting will be done to minimize noise and ground vibration with permission from DGMS shall be taken and shall be implemented under the control of mines manager

(Total 1000 characters)

9.3.7: Socio-Economic Environment

9.3.7.1: Demographic Profile

| S.N. | Type of Area | Name of Village | Total Population | Male to Female Ratio | Literacy Rate (%) | Employment Rate (%) |
|------|--------------|-----------------|------------------|----------------------|-------------------|---------------------|
|      |              |                 |                  |                      |                   |                     |



|   |             |                  |         |       |       |       |
|---|-------------|------------------|---------|-------|-------|-------|
| 1 | Residential | Haripura         | 1237.00 | 1.11  | 58.73 | 51.86 |
| 2 | Residential | Bairas           | 1237.00 | 1.003 | 62.11 | 58.60 |
| 3 | Residential | Thoryon Ki Dhani | 709.00  | 1.00  | 60.49 | 55.54 |
| 4 | Residential | Khorwa           | 1446.00 | 1.00  | 44.26 | 52.38 |
| 5 | Residential | Jorawarpura      | 729.00  | 1.03  | 26.55 | 50.65 |
| 6 | Residential | Dehro            | 2762.00 | 0.99  | 57.45 | 67.35 |

9.3.7.1.1: Anticipated Impact

(Give details about impact on the cropping pattern & crop productivity in the core zone)

In next five year mining shall be done on Government and private land so there is impact on crop production Within private land (2.1544 ha). During the final stages of mining in the area crop productivity shall be reduced as mining activity will be carried out over Private land and government land.

(Total 1000 characters)

9.3.7.1.2: Mitigation Measure

(Give details about compensation for loss of land & crops)

1). Land owners will be given compensation for their lands.2). Some amount shall be spent for fauna. 3). Jobs will be given to the families of land owners nearby for their families and livelihoods.

(Total 1000 characters)

9.3.7.2: Traditional Skills & Source of Livelihood

9.3.7.2.1: Base / Present Status

(Give details about present status on traditional skills & source of livelihood)

Presently the main source of living is farming and wage labour.

(Total 1000 characters)

9.3.7.2.2: Anticipated Impact

(Give details about positive & negative impacts on present status of livelihood in the area)

Adverse Social impacts: Price Inflation which is increase in purchasing power of the community members through higher incomes from the mining, compounded by population increase and low agricultural productivity as a result of project in cluster will lead to inflation in the cost of goods and services. Livelihood change due to the labor intensity of the mining, the project in cluster will attract the more able-bodied persons from the community which in turn will lead to low labour availability in other sectors of the economy including agricultural, education and health skilled workers. Local employment opportunities will be created by the project. This impact will not be significant due to low level of education and skills in the area which will result in sourcing skilled work force from out side the immediate area. But the magnitude of this impact will be high due to high number of dependents in a household. Positive impacts:1)Employment: Employment will be created during planning and preparation, construction and operational phases of the project. Employment opportunities created by lessees will provide a sustainable and safeworking environment for women. 2) CommunitySkills Development: The employees will be nefit from the training programmers that will be instituted by leases to enable the community labor forceto work in the different areas of project operations.This training will increase the number of technicians, electricians, and mechanics, among others,that will not only benefit of leases but also the community at large during and after the project life.3) Improved Standard of Living: Employment opportunities created by the projects incluster will increase income and therefore improve the overall standards of living in the area. 4) Community Organizational Capacit Development : Through engagement of community members in development structures such as Community Development Arena.

(Total 1000 characters)

**9.3.7.2.3: Mitigation Measure**

(Give details about training to locals for employment in the project training for making them self-employable or elsewhere)

Training will be given to locals for- 1 females for stitching work amrodry work 2 training for technicians,electricians,mechanics driving, HEMM operators and other activities to make them self-employable.

(Total 1000 characters)

**9.3.7.3: Economic Profile of the Population in Core & Buffer Zone**

**9.3.7.3.1: Base / Present Status**

(Give details about economic profile of the population in core & buffer zone)

The villagers in the core zone and buffer zone are poor and mainly dependent on farming cattle grazing animal husbandry and small grocery shop

(Total 1000 characters)

**9.3.7.3.2: Anticipated Impact**

(Give details about impact on community resources such as grazing land)

No grazing land in the lease area so there is no impact on grazing land

(Total 1000 characters)

**9.3.7.3.3: Mitigation Measure**

(Give details about employment opportunities & access to other amenities such as education, health care facilities to be extended to locals, addressing local unemployment, tourism or recreation opportunities, efforts for sustainable development of the local community)

About 6 to 10 people will get direct employment and about 50 people will get indirect employment the lessee shall give rupees 25 thousand per year funds to improve eduction facilities to nearby schools the lese shall contribute funds of rupees 25 thousand to organize health are camps and help in providing health care facilities to locals the lesse shall also provide rupees 50 thousand fund for village development

(Total 1000 characters)

**9.3.7.4: Human Settlement in Core & Buffer Zone**

**9.3.7.4.1: Base / Present Status**

(Give details about human settlement in core & buffer zone)

The lease block does cover three huts Hence the mining activity in the area will involve displacement of human settlement No village falls within the block The mining operations will not disturb or relocate any village Three families has to displacement from the block

(Total 1000 characters)

**9.3.7.4.2: Anticipated Impact**

(Give details about any displacement of human settlements during the life of the mine)

There are three hut with in the block so displacement of families is anticipated

(Total 1000 characters)

**9.3.7.4.3: Mitigation Measure**

(Give details about rehabilitation & resettlement of land ousters & displaced people)

There are three families exist in the block which shall be displaced by giving funds and jobs to them

(Total 1000 characters)

9.3.7.5: Health Profile of Population in Core & Buffer Zone

9.3.7.5.1: Base / Present Status

(Give details about health profile of population in core & buffer zone)

The data reveals that the area is lacking in medical facilities as 3 villages in the core area has medical facility There are 3 Government health centers 1 Ayurvedic dispensaries and only11 Primary health centre in the region The region is very poor in medical facilities even after the introduction of Centrally Sponsored NRHM Program in the state Medical Facilities In and near core area 3 villages are present with 2 medical facilities and 2primary health sub centre In buffer area

(Total 1000 characters)

9.3.7.5.2: Anticipated Impact

(Give details about any adverse impact on the general health condition of the population in core & buffer zone)

The projects will not impact the human health The most common diseases in the region is common cold and cough malaria flue enteric diseases like typhoid which will be prevalent amongst the mine workers None of these can be related to the project allied activities in the area The quality of water both surface and ground water is good and all parameters of drinking water are as per standards Wate ranalysis will be carried out at periodical intervals There fore the water will not cause any problem in the buffer zone The scope of any disease due to dust does not arise the emission monitored is below the prescribed limit and will get confined to mining area and will get arrested by green belt all around the mine Noise generation will be kept low and can not travel beyond the block area Therefore no disease in the impact zone can be related to project or any activity of the project Hence question of remedial measures does not arise

(Total 1000 characters)

9.3.7.5.3: Mitigation Measure

(Give details about avenues like dispensaries, hospitals, maternity homes if any to be created)

A fund of rupees 25 thousand will be given to a near by dispensary for up gradation of medical facilities

(Total 1000 characters)

9.3.7.6: Historically, Culturally & Ecologically Important Places in Core & Buffer Zone

9.3.7.6.1: Base / Present Status

(Give details about historically, culturally & ecologically important places in core & buffer zone)

In core area no historically culturally important places and Ecologically important places are present In buffer area no historically culturally important places and Ecologically important places are

present

(Total 1000 characters)

9.3.7.6.2: Anticipated Impact

(Give details about risk profiling)

No Impact as no important places are present

(Total 1000 characters)

9.3.7.6.3: Mitigation Measure

(Give details about public health benefits (e.g. clean water to an aboriginal community), measure for safeguard against damage etc.)

Not required due to above stated reasons

(Total 1000 characters)

Approved

## Chapter 10: Annexures

### 1. Upload Document

#### 1.1 Upload Document

| Sl.No. | Title                                                                              | Is Upload | Document (only pdf allowed)                             |
|--------|------------------------------------------------------------------------------------|-----------|---------------------------------------------------------|
| 1      | Letter of Intent /Letter of lease grant                                            | Nil       | <a href="#">Block 18 LOI.pdf</a>                        |
| 2      | Copy of lease deed executed                                                        | Nil       | Nil                                                     |
| 3      | Copy of Declaration of Owner/Nominated Owner in case of Company/partnership firm   | Nil       | <a href="#">POA with affidavit and BR.pdf</a>           |
| 4      | ID & Address Proof of Owner/ Nominated Owner                                       | Nil       | <a href="#">Aadhar card- Mehul Agarwal.pdf</a>          |
| 5      | Copy of Environment and Forest Clearence, Consent to Establish, Consent to Operate | Nil       | <a href="#">Undertaking.pdf</a>                         |
| 6      | Copy of Registration of Company (RoC)/Partnership firm (Registration) & Deed       | Nil       | <a href="#">Common COI SUNCHEM SUNMET GST.pdf</a>       |
| 7      | Consent letter for Qualified Person                                                | Nil       | <a href="#">Consent Letter-Lease code PB216 new.pdf</a> |
| 8      | Experience & Qualification Details of Qualified Person                             | Nil       | <a href="#">Degree.pdf</a>                              |
| 9      | Certificate from QP                                                                | Nil       | <a href="#">Certificate from QP.pdf</a>                 |
| 10     | Copy of Bank Guarantee                                                             | Nil       | <a href="#">BG-Block 18 (1).pdf</a>                     |
| 11     | Copy of Performance Surety                                                         | Nil       | <a href="#">BG-Block 18 (1).pdf</a>                     |
| 12     | Copy of MDPA (as applicable)                                                       | Nil       | Nil                                                     |
| 13     | Exploration details                                                                | Nil       | <a href="#">Exploration.pdf</a>                         |
| 14     | Copy of feasibility Report                                                         | Nil       | <a href="#">NEW complet PFR 18.pdf</a>                  |
| 15     | Copy of Study reports conducted as per Para                                        | Nil       | Nil                                                     |



|    |                                                                          |     |                                                |
|----|--------------------------------------------------------------------------|-----|------------------------------------------------|
|    | 4.3.1                                                                    |     |                                                |
| 16 | Chemical and Mineralogical analysis report                               | Nil | <a href="#">Chemical Analysis.pdf</a>          |
| 17 | Any other Report or Certification as required in the submitted Document. | Nil | Nil                                            |
| 18 | Copy of Scale relaxation approval granted(if applicable)                 | Nil | Nil                                            |
| 19 | Mineral processing flowsheet with stage wise recovery                    | Nil | Nil                                            |
| 20 | Any Other (1)                                                            | Yes | <a href="#">Monitoring_and_undertaking.pdf</a> |
| 21 | Any Other (2)                                                            | Yes | <a href="#">Final_new_annexure.pdf</a>         |

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## Chapter 11: Plates (OC)

### 1. Upload Document

#### 1.1 Upload Document

| S.N. | Title                                                                                                                                                                                                                                                                                                                                                                                                             | Is Upload | Document                                                |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|---------------------------------------------------------|
| 1    | Lease sketch plan;                                                                                                                                                                                                                                                                                                                                                                                                | Nil       | <a href="#">Lease sketch plan.pdf</a>                   |
| 2    | Surface Plan (.KMZ format)(Georeferenced); A statutory plan as per MCDR, 2017. The Plan should be submitted showing different color codes for:(1) Active Pits & Excavation area(2) Excavated area reclaimed & rehabilitated (3)Active dumps (4) Stabilized & rehabilitated dump area , (5) Green belt (6) Mineral Stacks (7) Utilities such as plant, buildings etc (8) Lease boundary along with other details.) | Nil       | <a href="#">Surface_Plan.kmz</a>                        |
| 3    | Surface Geological Plan of the lease (.KMZ format)(Georeferenced); The Plan should be submitted showing different color codes for : (1) Lithological/Geological Occurance (2) Area under G1,G2,G3 & G4 (3) Active pits & Excavation area (4) Dump Area (5) Mineral Stacks (6) Lease boundary along with other details.)                                                                                           | Nil       | <a href="#">Surface Geological Plan.kmz</a>             |
| 4    | Surface Geological sections (in Pdf format); Geological sections with different color coding depicting all the features shown in Surface Geological Plan. )                                                                                                                                                                                                                                                       | Nil       | <a href="#">Geological_Plan_And_Section.pdf</a>         |
| 5    | Five year Production and Development plan (.KMZ format)(Georeferenced); The Plan should be submitted showing different color coding for: (1) Active Pit and Excavation area ,                                                                                                                                                                                                                                     | Nil       | <a href="#">Proposed_Five_Year_Development_Plan.kmz</a> |



|    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |     |                                                   |
|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|---------------------------------------------------|
|    | (2) Year wise excavation proposal for year I to V (3) Active dump and yearwise dump proposal for year I to V (4) Year wise Dump working proposal for year I to V (6) Lease boundary (with reference to chapter 4) along with other details.)                                                                                                                                                                                                                                                    |     |                                                   |
| 6  | Five year Production and Development sections (in pdf format); Year wise excavation and dumping proposals with different color coding depicting all the features as shown in the Five year Production and development plan.)                                                                                                                                                                                                                                                                    | Nil | <a href="#">Five_year_production.pdf</a>          |
| 7  | Progressive Mine Closure Plan (.KMZ format)(Georeferenced); The Plan should be submitted showing different color coding for :<br>(1) Yearwise excavated area Reclaimed & rehabilitated for year I to V (2) Year wise dump area to be stabilized and dump area to be rehabilitatd for year I to V (3) Year wise Green area proposed from year I to V.(4) Any other reclamation and rehabilitation measures proposed.(5) Lease boundary ( with reference to chapter 6) along with other details.) | Nil | <a href="#">Progressive_Mine_Closure_Plan.kmz</a> |
| 8  | Progressive mine Closure sections (in pdf format); Year wise Progressive mine clouser sections showing all the yearwise reclamation, rehabilitaion proposals as depicted in the Progessive mine clouser plan.)                                                                                                                                                                                                                                                                                  | Nil | <a href="#">Progressive_Mine_clousar_Plan.pdf</a> |
| 9  | Conceptual Plan (.KMZ format)(Georeferenced); The Plan should depict the staus of lease area as envisaged at the end of life of Mine showing all the details. Status of land use shall be depicted by different color coding.)                                                                                                                                                                                                                                                                  | Nil | <a href="#">Conceptual_Plan.kmz</a>               |
| 10 | Conceptual Sections (pdf) format;                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Nil | <a href="#">Conseptual_plan.pdf</a>               |
| 11 | Geo referenced Cadastral Plan; Duly certified by the State Government)                                                                                                                                                                                                                                                                                                                                                                                                                          | Nil | <a href="#">CADASTRAL_PLAN.pdf</a>                |
| 12 | Financial Assurance Plan (KMZ);                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Nil | <a href="#">Financial_Assurance_Area_Plan.kmz</a> |

|    |                                                                                                                                                                                                                  |     |                                             |
|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|---------------------------------------------|
| 13 | Environmental Plan (.KMZ format)(Georeferenced); As per MCDR, 2017 indicating all the details.)                                                                                                                  | Nil | <a href="#">Enviromental_Plan.kmz</a>       |
| 14 | Any other plan/section as deemed necessary by approving authority;                                                                                                                                               | Yes | <a href="#">Environment_or_key_plan.pdf</a> |
| 15 | Five Year Production and Development sections (in pdf format);                                                                                                                                                   | Yes | <a href="#">Fifth_Year_production.pdf</a>   |
| 16 | LEVEL WISE SLICE PLAN; LEVEL WISE SLICE PLAN (PDF FORMAT IN VISIBLE SCALE))                                                                                                                                      | Nil | Nil                                         |
| 17 | Zipped shp file of the lease boundary; Upload a single zipped file of the lease boundary)                                                                                                                        | Nil | <a href="#">Lease_Boundary.zip</a>          |
| 18 | Zipped shp file of the boundary pillars; Enter a single zipped file of the boundary pillars.)                                                                                                                    | Nil | <a href="#">Lease_Pillar.zip</a>            |
| 19 | Zipped shp file of the Production and Development plan for respective proposal year - Year1; Upload a zipped shape file indicating a separate production and development plan for each respective proposal year) | Nil | <a href="#">First_Year_Pit.zip</a>          |
| 20 | Zipped shp file of the Production and Development plan for respective proposal year - Year2; Upload a zipped shape file indicating a separate production and development plan for each respective proposal year) | Nil | <a href="#">Second_Year_Pit.zip</a>         |
| 21 | Zipped shp file of the Production and Development plan for respective proposal year - Year3; Upload a zipped shape file indicating a separate production and development plan for each respective proposal year) | Nil | <a href="#">Third_Year_Pit.zip</a>          |
| 22 | Zipped shp file of the Production and Development plan for respective proposal year - Year4; Upload a zipped shape file indicating a separate production and development plan for each respective proposal year) | Nil | <a href="#">Fourth_Year_Pit.zip</a>         |
| 23 | Zipped shp file of the Production and                                                                                                                                                                            | Nil | <a href="#">Fifth_Year_Pit.zip</a>          |

|    |                                                                                                                                                                            |     |                                      |
|----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|--------------------------------------|
|    | Development plan for respective proposal year - Year5; Upload a zipped shape file indicating a separate production and development plan for each respective proposal year) |     |                                      |
| 24 | Zipped shp file of the Dump plan for respective proposal year - Year1; Upload zipped file of the dump plan separately for each respective proposal year)                   | Nil | <a href="#">First_Year_Dump.zip</a>  |
| 25 | Zipped shp file of the Dump plan for respective proposal year - Year2; Upload zipped file of the dump plan separately for each respective proposal year)                   | Nil | <a href="#">Second_Year_Dump.zip</a> |
| 26 | Zipped shp file of the Dump plan for respective proposal year - Year3; Upload zipped file of the dump plan separately for each respective proposal year)                   | Nil | <a href="#">Third_Year_Dump.zip</a>  |
| 27 | Zipped shp file of the Dump plan for respective proposal year - Year4; Upload zipped file of the dump plan separately for each respective proposal year)                   | Nil | <a href="#">Fourth_Year_Dump.zip</a> |
| 28 | Zipped shp file of the Dump plan for respective proposal year - Year5; Upload zipped file of the dump plan separately for each respective proposal year)                   | Nil | Nil                                  |

# Chapter 11 : Plates(UG) : NA

Approved



Approved